



Operating Systems & Networks

CTEC1704C

Lecture – 10 : The Application Layer

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Lecture Content

- Application Layer
 - Architectures
 - Addressing
 - Transport service requirements
- The Telnet Protocols
- Hypertext Transfer Protocol



References & Reading

- Kurose & Ross; Computer Networking;
 - Chapter 2
- http://www.tcpipguide.com/free/t_TelnetProtocol.htm
- <https://en.wikipedia.org/wiki/HTTP/>
- [2https://secure.wikimedia.org/wikipedia/en/wiki/List_of_HTTP_header_fields](https://secure.wikimedia.org/wikipedia/en/wiki/List_of_HTTP_header_fields)
- <https://www.w3.org/Protocols/rfc2616/rfc2616-sec10.html>



Post lecture discussion topics

- What protocol is discussed in RFC 1?
 - Briefly explain each of the header fields
- Investigate Telnet security vulnerabilities
 - what are the issues and how are they resolved?
- Distinguish between safe and idempotent HTTP request methods
 - Make a list of methods under each heading

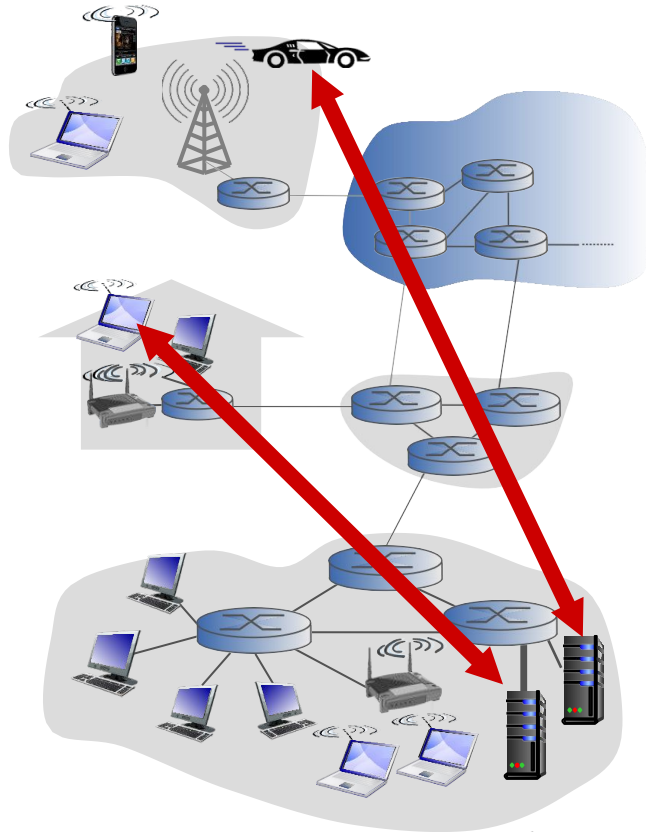


Lecture Content

- **Application Layer**
 - Architectures
 - Addressing
 - Transport service requirements
- The Telnet Protocols
- Hypertext Transfer Protocol

Client-server architecture

One-to-one
asynchronous
relationship





Client-server architecture

- Server:
 - always-on host
 - physical & process
 - permanent IP address
 - data centres for scaling

Client-server architecture

- Clients:
 - communicate with server
 - may be intermittently connected
 - may have dynamic IP addresses
 - do not communicate directly with each other



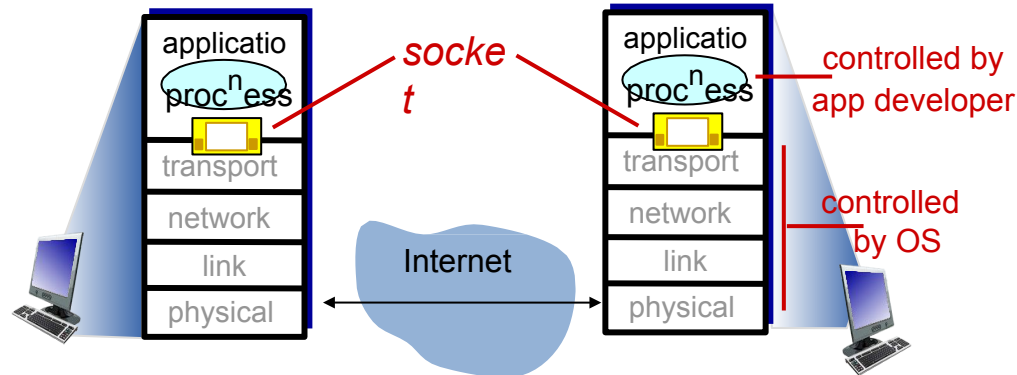
Accessing a process

- To receive messages, server process must have identifier
- Host device has globally unique 32-bit IP address
 - IP address of host on which process runs does not suffice for identifying the process because many processes can be running on same host
- Identifier includes both **IP address** and **port numbers** associated with process on host
- Example port numbers:
 - HTTPS server: 443
 - POP mail server: 110



Sockets

- Process sends/receives messages to/from its socket
- Each socket corresponds to one port number
- IP address and port number are bound
- Sending process relies on transport infrastructure to deliver message to socket at receiving process



App-layer protocol defines

- Types of messages exchanged
 - eg request, response
- Message syntax:
 - what fields in messages & how fields are delineated
- Message semantics
 - meaning of information in fields
- Rules for when and how processes send & respond to messages

App-layer protocol types

- Open protocols:
 - defined in RFCs
 - allows for interoperability
 - eg HTTP,
SMTP
- Proprietary protocols:
 - eg Skype



What transport service do apps need?

- data integrity
 - some apps (eg file transfer, web transactions) require 100% reliable data transfer
 - other apps (eg audio) can tolerate some loss
- timing
 - some apps (eg Internet telephony, interactive games) require low delay to be “effective”



What transport service do apps need?

- throughput
 - some apps (eg multimedia) require minimum amount of throughput to be “effective”
 - other apps (“elastic apps”) make use of whatever throughput they get
- Security
- encryption, data integrity, ...



Transport service requirements: common apps

application	data loss	throughput	time sensitive
file transfer	no loss	elastic	no
e-mail	no loss	elastic	no
Web documents	no loss	elastic	no
real-time audio/video	loss-tolerant	audio: 5kbps - 1Mbps video: 10kbps - 5Mbps	yes, 100's msec
stored audio/video	loss-tolerant	same as above	yes, few secs
interactive games	loss-tolerant	few kbps up	yes, 100's msec
text messaging	no loss	elastic	yes and no

Lecture Content

- Application Layer
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- **The Telnet Protocols**
- Hypertext Transfer Protocol



- . TELetype NETwork
- . RFC 15 (1969)
 - <https://www.ietf.org/rfc/rfc854.txt>
- Developed to solve the problem of one (local) terminal per computer
- Solution:
 - build a software stack divided into
 - . application
 - . network

ICL System 4/70



https://farm7.staticflickr.com/6215/6350917248_52137fe1bc_z.jp

9

Teletype



Just for comparison...



- Network Virtual Terminal
 - Implements client/server
- Requires a common message format

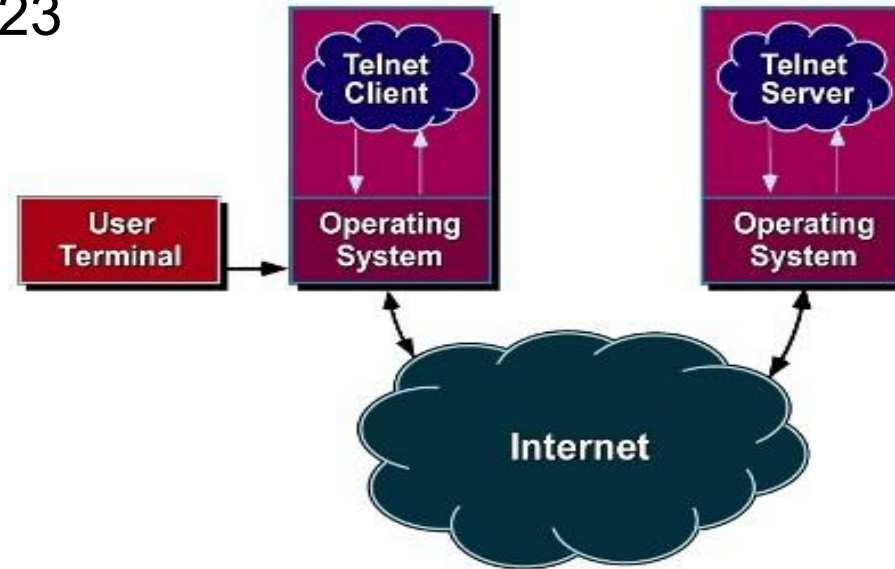
```
|<----- 16 ---->|<-8->|<-8->|<-- 16 -->|<-8->|<---
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
|   leader           |  x  |size |  count   |  x  | TEXT
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
|<---- level 1 ---->|
      message leader

|<----- level 2 ----->|
      message preamble

                                level 3
                                |<- message text..-->
```

Telnet

- 1 byte per character
- Full duplex – listens on port 23



Telnet problems

- Messages are not encrypted
 - plain text
 - packet sniffing allows easy detection of passwords
- No authentication system
 - extremely vulnerable to M-I-T-M attacks
 - protocols such as FTP may use a weak authentication
- Many exploitable vulnerabilities have been discovered in Telnet servers

Other Telnet Protocols

- Many standard Internet protocols are based upon the concepts of this protocol
 - FTP, HTTP, SMTP, etc.
- They suffer from many of the same problems

Lecture Content

- Application Layer
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- The Telnet Protocols
- **Hypertext Transfer Protocol**

- HTTP



Web and HTTP

- First, a review...
- web page consists of objects
- object can be HTML file, JPEG image, applet, audio file,...
- web page consists of base HTML-file which includes several referenced objects
- each object is addressable by a URL

www.someuni.ac.uk/someDept/pic.gif

host name

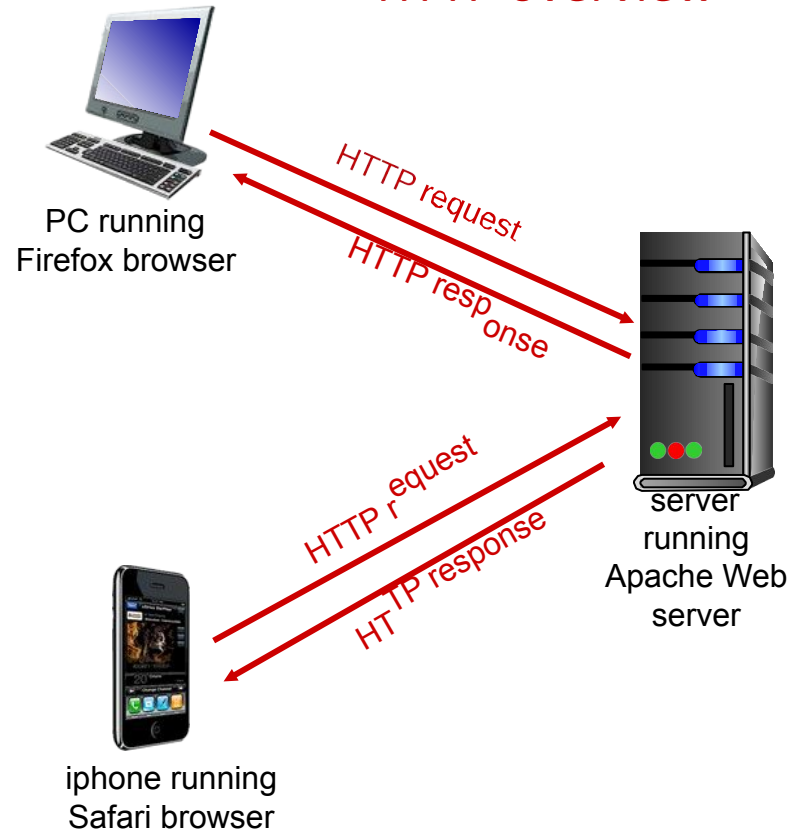
path name



HTTP overview

- Web's application layer protocol
- Client/server model
 - client: browser that requests, receives, (using HTTP protocol) and “renders/displays” Web objects
 - server: Web server sends (using HTTP protocol) objects in response to requests
- Uses TCP
 - server listens for client connections on port 80
- Is stateless
 - server does not maintain information about past client requests

HTTP overview

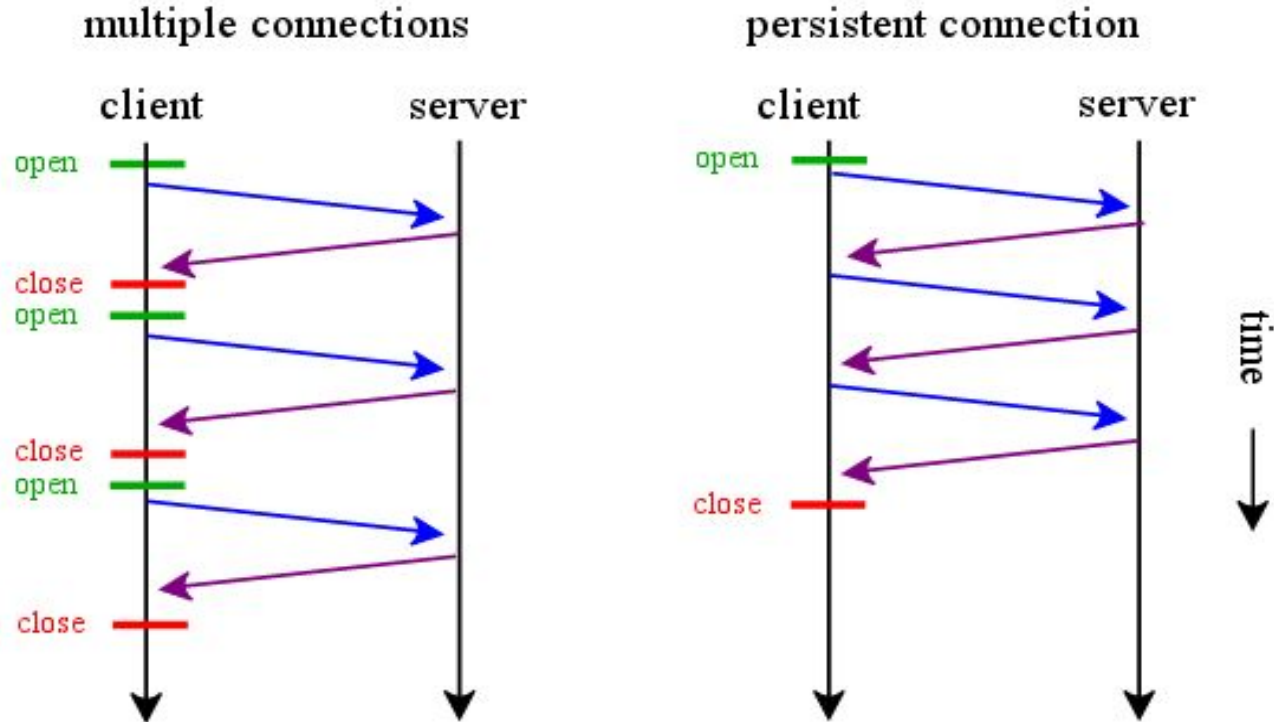




HTTP versions

- 1.0 = non-persistent connections
 - at most one object sent over TCP connection
 - connection then closed
 - downloading multiple objects requires multiple connections
- 1.1 = persistent connections
 - multiple objects can be sent over single TCP connection
 - pipelining
- 2.0 = Push technology
- 3.0 = QUIC technology – runs over UDP

HTTP versions



HTTP Messages

- Two types of HTTP messages
 - request, response
- HTTP messages use characters sets such as UTF-8
 - human-readable format



HTTP request message

request line
(GET, POST, commands)

carriage return character

line-feed character

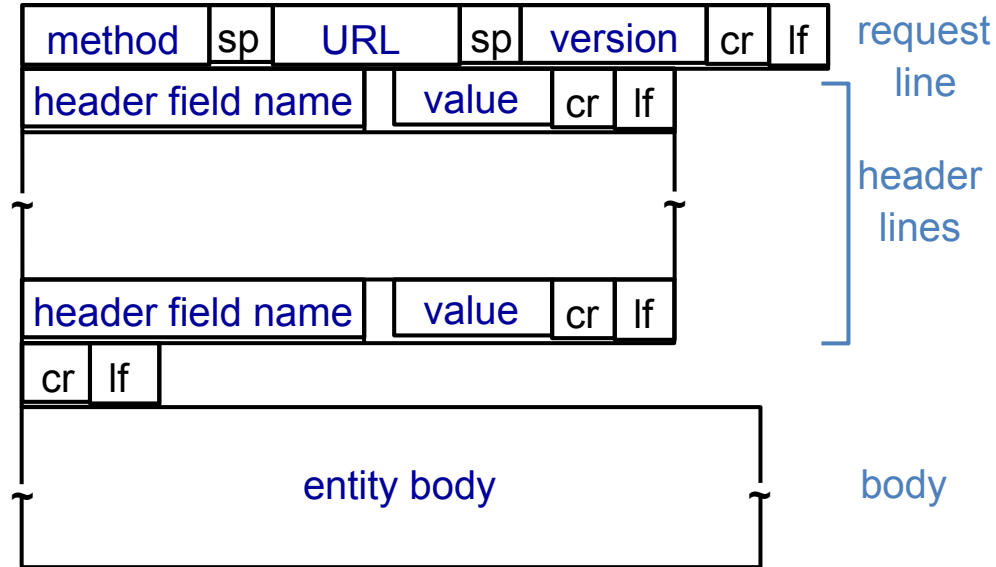
header lines

```
GET /index.html HTTP/1.1\r\n
Host: www-net.cs.umass.edu\r\n
User-Agent: Firefox/3.6.10\r\n
Accept: text/html,application/xhtml+xml\r\n
Accept-Language: en-us,en;q=0.5\r\n
Accept-Encoding: gzip,deflate\r\n
Accept-Charset: ISO-8859-1,utf-8;q=0.7\r\n
Keep-Alive: 115\r\n
Connection: keep-alive\r\n
\r\n
```

carriage return,
line feed at start
of line indicates
end of header lines

HTTP request message:

general format



HTTP response message

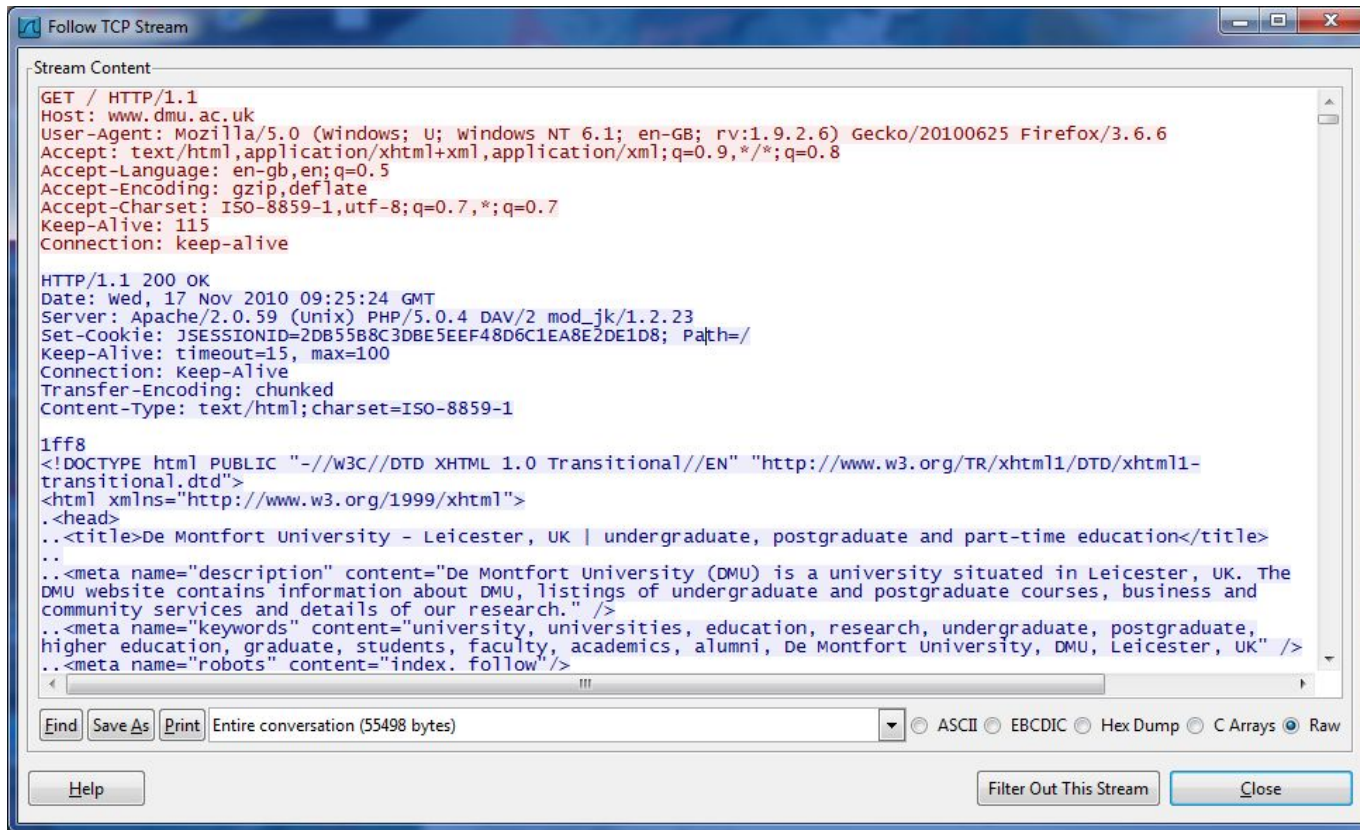
status line
(protocol
status code
status phrase)

header
lines

data, e.g.,
requested
HTML file

```
HTTP/1.1 200 OK\r\n
Date: Sun, 26 Sep 2010 20:09:20
GMT\r\n Server: Apache/2.0.52
(CentOS)\r\n
Etag: Modified: Tue, 30 Oct 2007
117606025c-bf716880"\r\n
Accept-Ranges: bytes\r\n
Keep-Alive: timeout=10,
max-keep-alive=252\r\n
Content-Length: 2652\r\n
Content-Type: text/html;
charset=ISO-8859-1\r\n
\r\n
data data data data data ...
```

Wireshark Analysis





HTTP response status codes

- Informational 1xx
 - Request received, continuing process
- Successful 2xx
 - Requested action was received, understood and processed
- Redirection 3xx
 - Redirection
- Client Error 4xx
 - The client (user) seems to have erred
- Server Error 5xx
 - server failed to fulfill a valid request

HTTP Request Methods

- Options, Get, Head, Post, Put, Delete, Trace, Connect
- Safe methods
 - Get & Head only retrieve data
- 'Unsafe' methods
 - Post, Put, Delete
- Trace
 - Usually turned off by default
 - can leak information
- Browsers only use GET & POST



Idempotent & Safe HTTP Methods

- A multiple request should have the same result as a single request
 - With no (unwanted) side effects

HTTP Method	Idempotent	Safe
OPTIONS	yes	yes
GET	yes	yes
HEAD	yes	yes
PUT	yes	no
POST	no	no
DELETE	yes	no
TRACE	yes	yes
CONNECT	no	no

<https://www.iana.org/assignments/http-methods>
http-methods.xhtml



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Thank You!

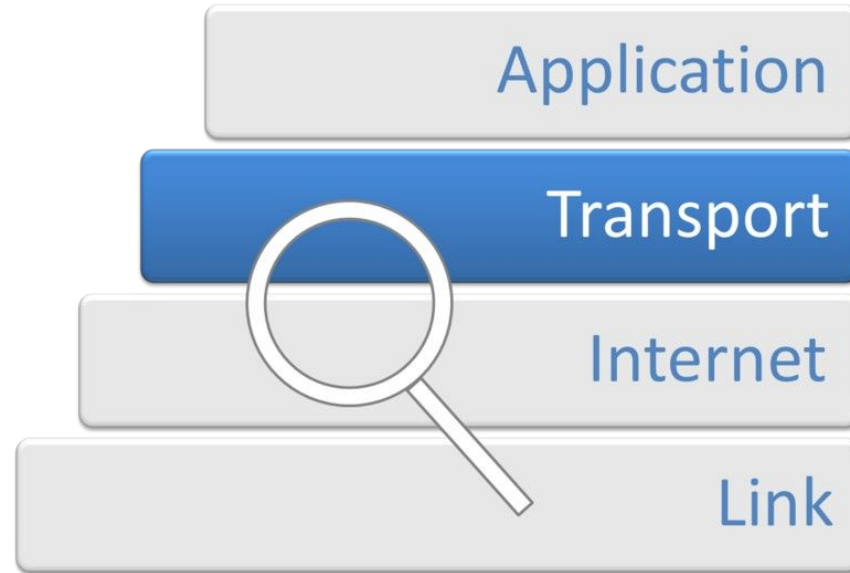


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Lecture – 11 : The Transport Layer

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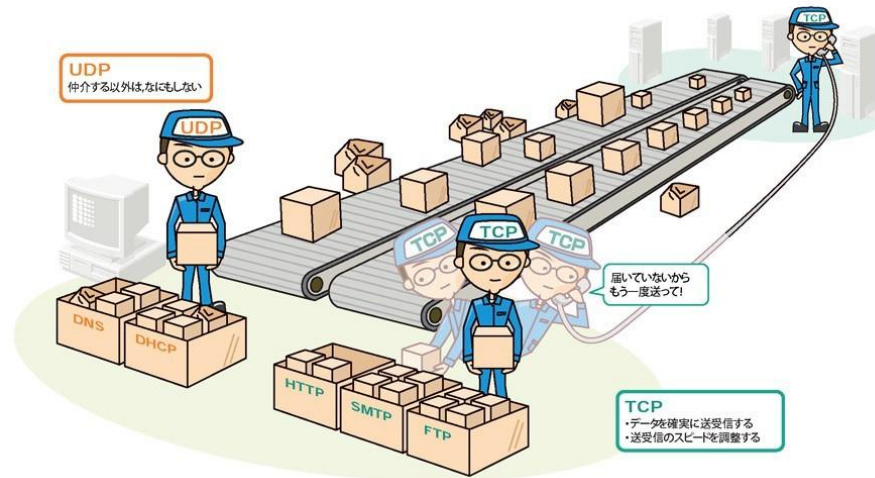


https://commons.wikimedia.org/wiki/File:Internet_Protocol_Analysis_-_Transport_Layer.png

The Transport Layer



- Principles of data transfer
- Transport Layer services Ping/ICMP
- UDP
- TCP





- Kurose & Ross; Computer Networking;
 - Chapter 3
- https://electronicspost.com/transport-layer-protocols-service_s_functions/
- <https://stackoverflow.com/questions/152457/what-is-the-difference-between-a-port-and-a-socket?rq=1>
- https://en.wikipedia.org/wiki/User_Datagram_Protocol
- https://en.wikipedia.org/wiki/Sliding_window_protocol
- https://en.wikipedia.org/wiki/Transmission_Control_Protocol

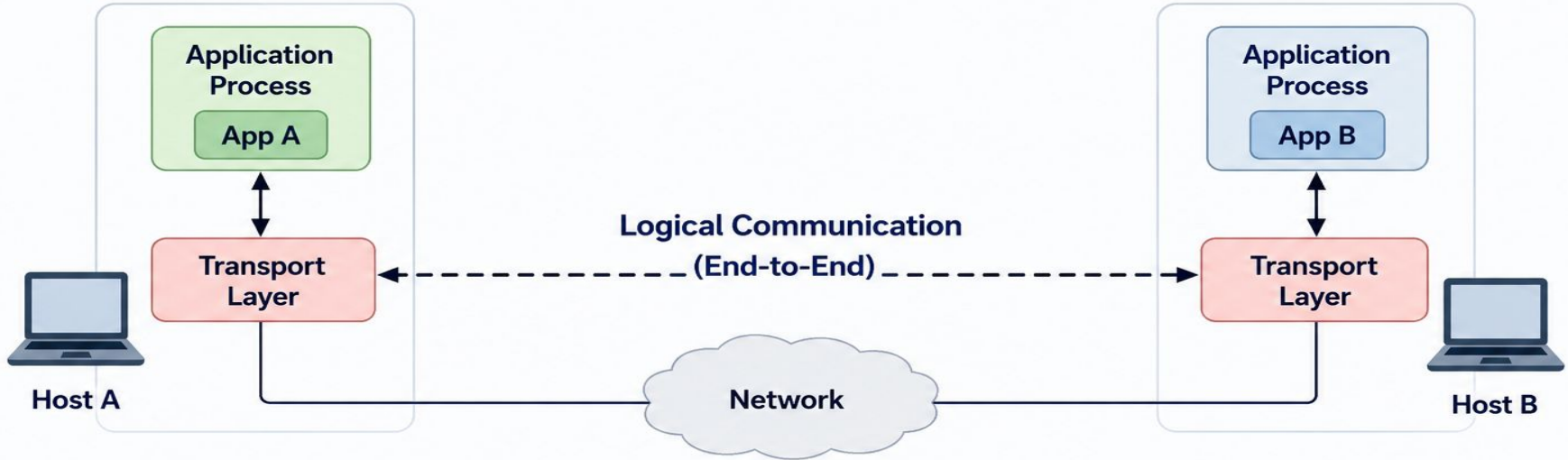


- Apart from predictable port numbers, what security issues can you find in the transport layer protocols?
- Read <https://www.ietf.org/rfc/rfc1149.txt>
 - Is this a protocol that will scale to cope with large amounts of data easily?

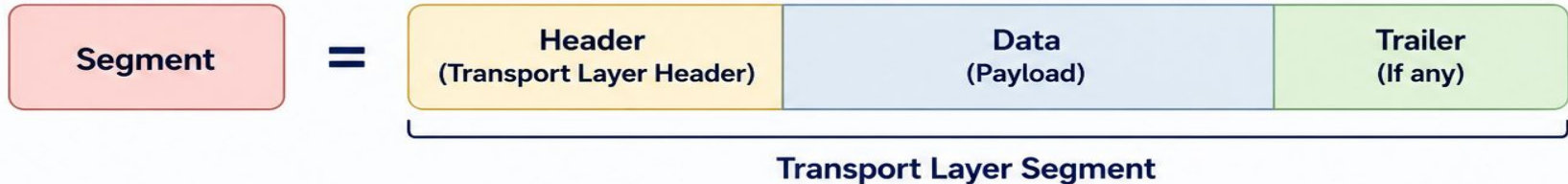


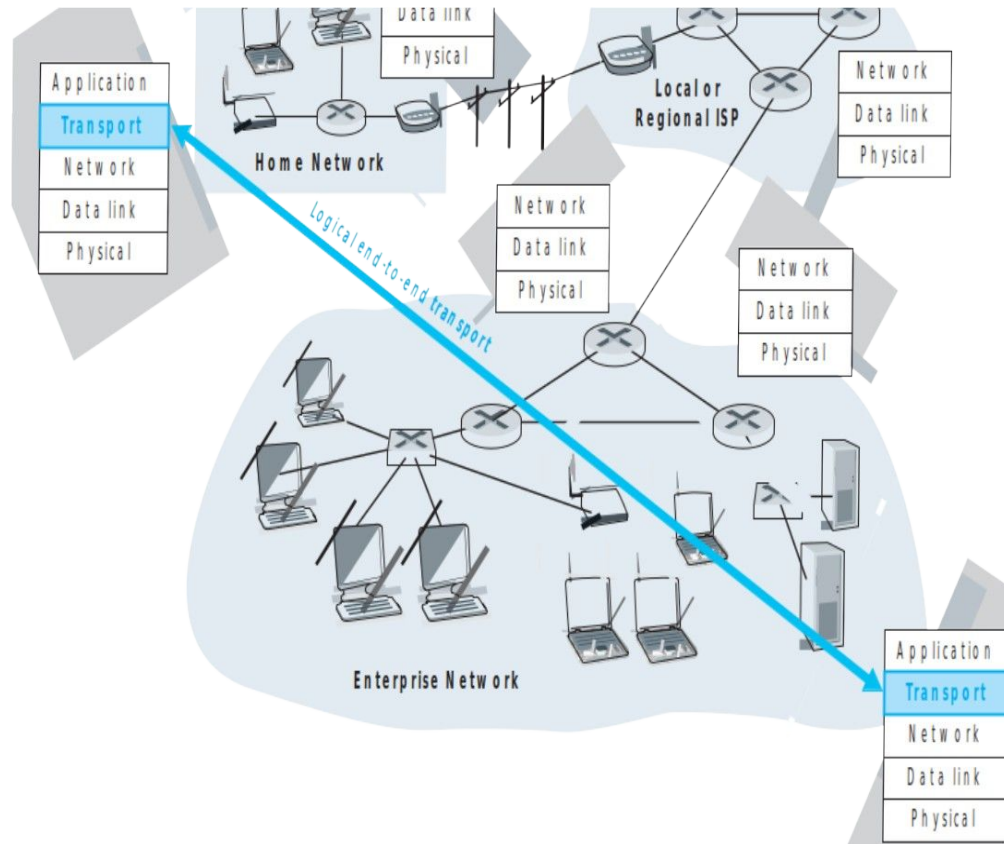
- Principles of data transfer
- Transport Layer services
- Ping/ICMP
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- TCP

The transport layer provides for logical communication between application processes



Transport layer packets are known as **segments**





Protocol port numbers



Port numbers identify the target processes at the application layer



16-bit from 0 – 65535



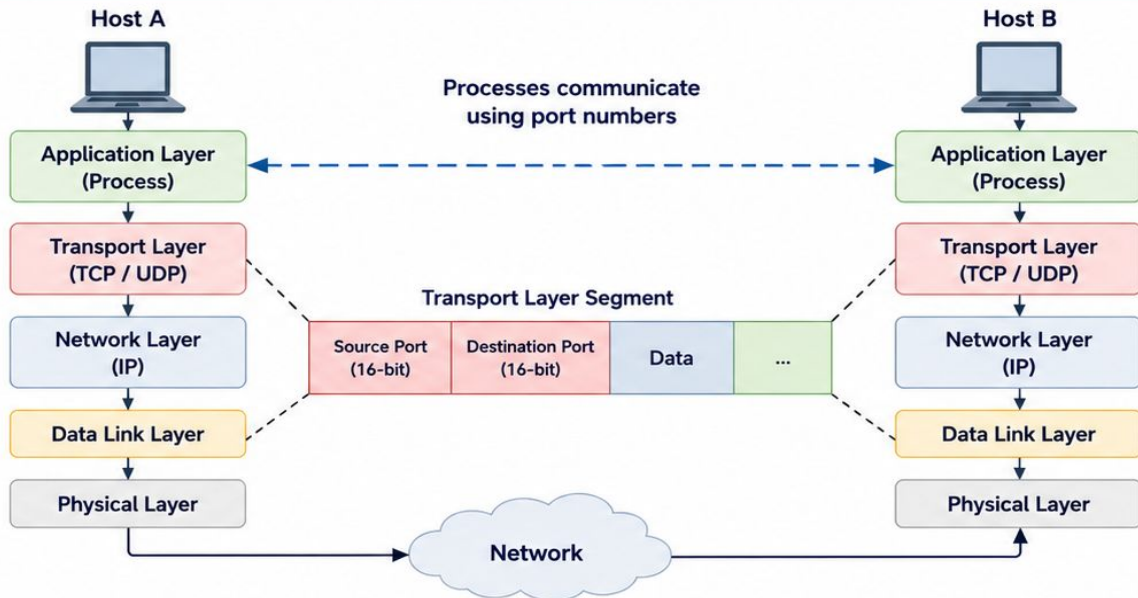
Both UDP and TCP



Allocated by IANA



RFC 1060, 1700 & 3232



Common Well-Known Port Numbers

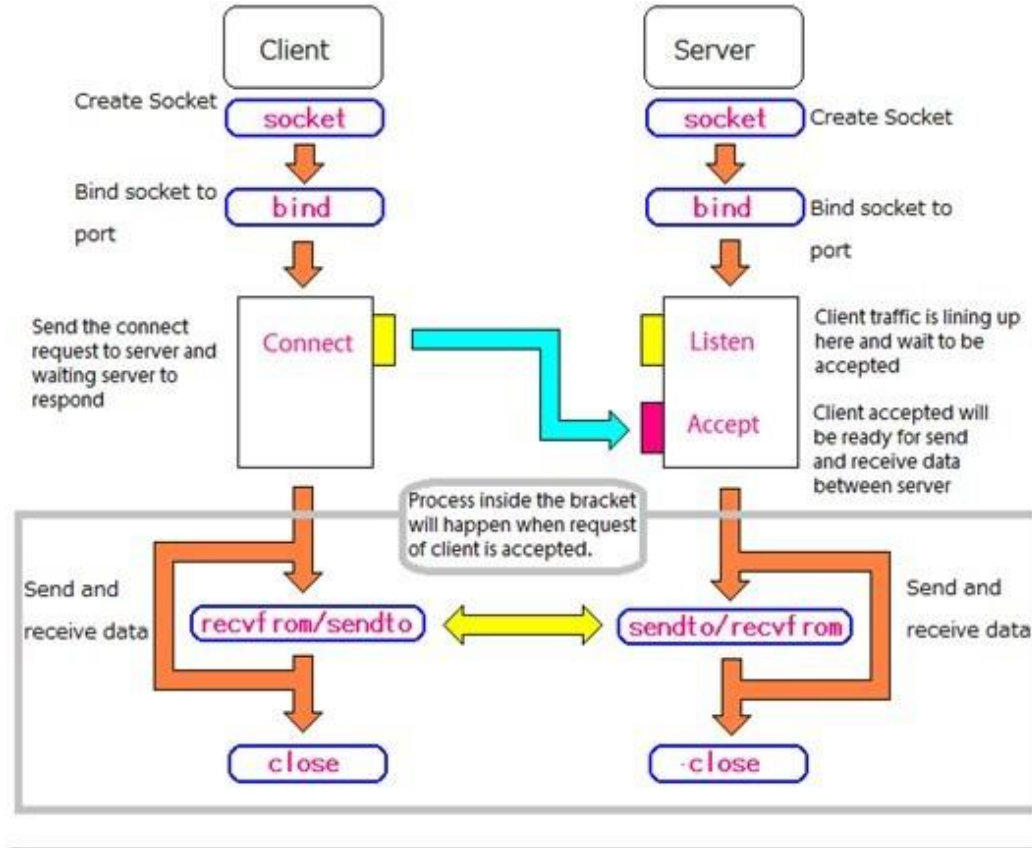
Port Number	Protocol	Service	Port Number	Protocol	Service
20	TCP	FTP (Data)	53	UDP/TCP	DNS
21	TCP	FTP (Control)	80	TCP	HTTP
22	TCP	SSH	110	TCP	POP3
25	TCP	SMTP	443	TCP	HTTPS



- Ports 0 - 1023
 - well-known port numbers
 - eg HTTP (80), FTP (21), SMTP (25)
- Ports 1024 - 4095
 - registered ports
 - eg MySQL/MariaDb (3306)
- Ports ≥ 4096
 - available to the OS



- Endpoints of a Transport Layer comms link
 - Combination of IP address and a port number
 - bound together
- A TL connection is defined by two endpoints
 - pair of sockets
 - client IP address & client port number
 - server IP address & server port
 - NOT the actual connection
- See
<https://stackoverflow.com/questions/152457/what-is-the-difference-between-a-port-and-a-socket?rq=1>



https://lh3.ggpht.com/-48jccDGoPY8/UYALpBdKml/AAAAAAAAAcr/JuqOmMNO_thumb%2525B4%2525D.png?imgmax=800



- Where does the *client* port number come from?
 - the operating system supplies a number
 - between 4096 and 65535
- Any security issue here?
 - predictable client port numbers...



- Principles of data transfer
- **Transport Layer services**
- UDP
- Ping/ICM
P
- TCP



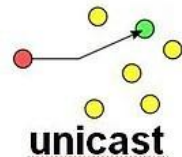
- In the Internet there are two main transport-layer services: **UDP** and **TCP**
 - UDP
 - User (Unreliable) Datagram Protocol
 - TCP
 - Transmission Control Protocol
- There is also
 - ICMP
 - Internet Control Message Protocol
 - ping, fping, traceroute, etc



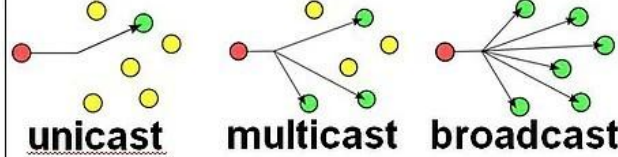
- UDP:
 - provides an unreliable, connectionless service to the application layer
 - does not guarantee that data sent by a processes at the application layer is arriving intact (or at all)
- TCP:
 - provides a reliable, connection-oriented services to the application layer



- **Slower but reliable transfers**
- **Typical applications:**
 - Email
 - Web browsing



- **Fast but non-guaranteed transfers ("best effort")**
- **Typical applications:**
 - VoIP
 - Music streaming



<https://knowledgeofthings.com/wp-content/uploads/2017/10/tcpvsudp.png>

- Must provide a reliable channel with protection against corrupted data and loss



- Principles of data transfer
- Transport Layer services
- UDP
- **Ping/ICMP**
- TCP

ICMP

- Internet Control Message Protocol (ICMP) (RFC 792)
- Used by hosts and routers to communicate network layer information
- Example: “Destination network unreachable” (ICMP Type 3)
- ICMP packet is encapsulated in an IPv4 datagram similar to TCP or UDP segments



ICMP: Message types and codes

ICMP Type	Code	Description
0	0	echo reply (to ping)
3	0	destination network unreachable
3	1	destination host unreachable
3	2	destination protocol unreachable
3	3	destination port unreachable
3	6	destination network unknown
3	7	destination host unknown
4	0	source quench (congestion control)
8	0	echo request
9	0	router advertisement
10	0	router discovery
11	0	TTL expired
12	0	IP header bad



- Ping
 - sends
 - ICMP type 8 code 0
 - to specified host
(request)
 - host replies
 - ICMP type 0 code 0
(echo)



- traceroute
 - is implemented using ICMP (type 11 code 0) and (type 3 code 3) and UDP
- Each packet is sent out with an incremented network layer TTL value
 - the router that the packet reaches when the TTL has reduced to zero discards the packet
 - sends an ICMP message to sender
 - sender resolves the router address



- Principles of data transfer
- Transport Layer services
- **UDP**
- Ping/ICMP
- TCP



- Why use UDP if TCP is offering so much more?
 - better application level control over data being sent
 - no need to establish connection
 - no connection state
 - smaller packets (UDP 8 byte vs TCP 20 byte)
 - speed!

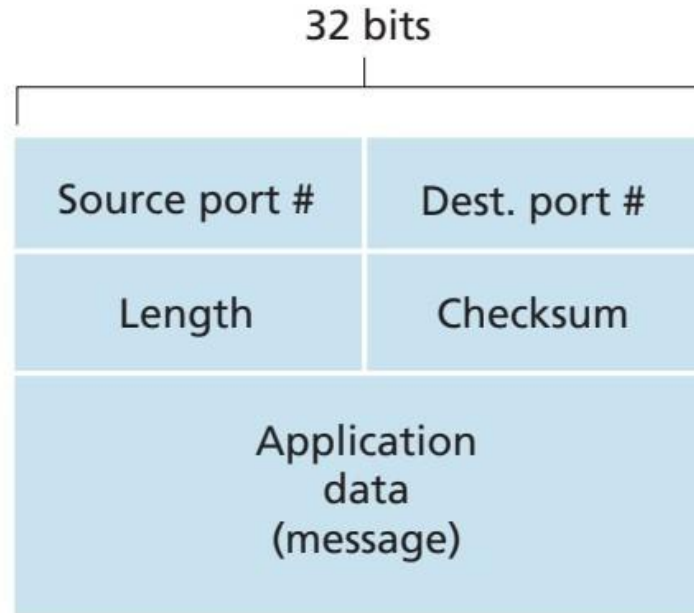


- Packet loss, reordering, errors or duplication are commonplace
 - remediation achieved at the application layer
 - . eg TFTP
 - may not be necessary
 - . eg VOIP
- Many key Internet applications use UDP
 - Domain Name System (DNS)
 - Simple Network Management Protocol (SNMP)
 - Routing Information Protocol (RIP)
 - Dynamic Host Configuration Protocol (DHCP)



- UDP Segment structure is defined in RFC768

- From RFC 768:
- The Source port # is optional
 - defaults to 0
- Similarly, the checksum is also optional
 - also defaults to 0

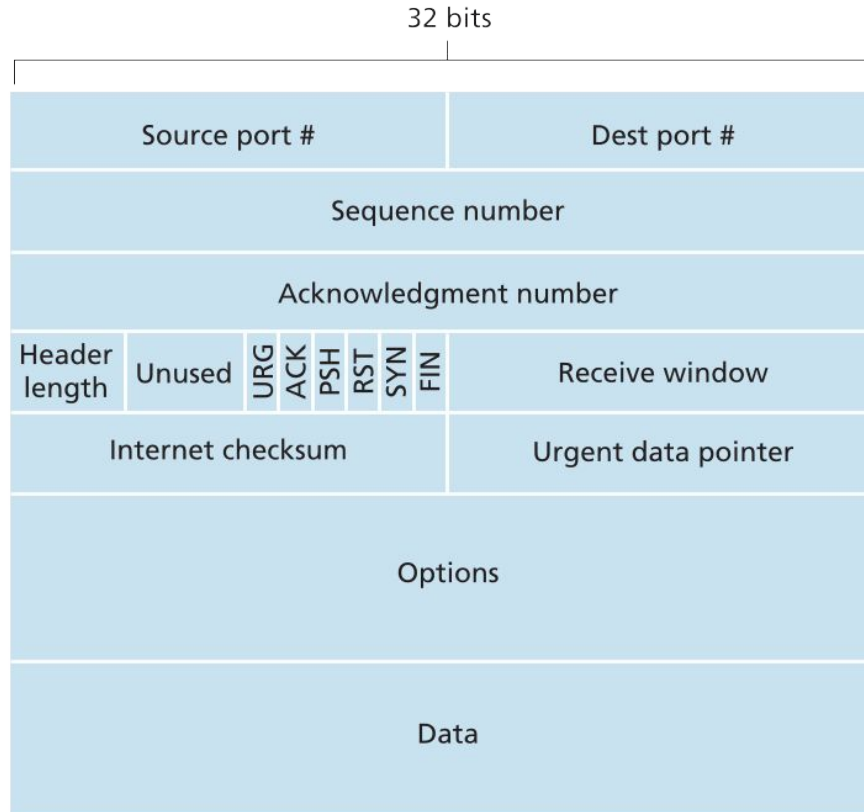




- Principles of data transfer
- Transport Layer services
- UDP
- Ping/ICMP
- **TCP**



- Creates reliable data transfer on top of IP's unreliable service
- Connection-Oriented Service
- Full Duplex
 - Data can flow both directions
- Point to Point
 - Single Sender and Single Receiver
- Set up during a three-way-handshake
 - connection state
 - buffers (Send and receive)
- Flow controlled: sender will not overwhelm receiver

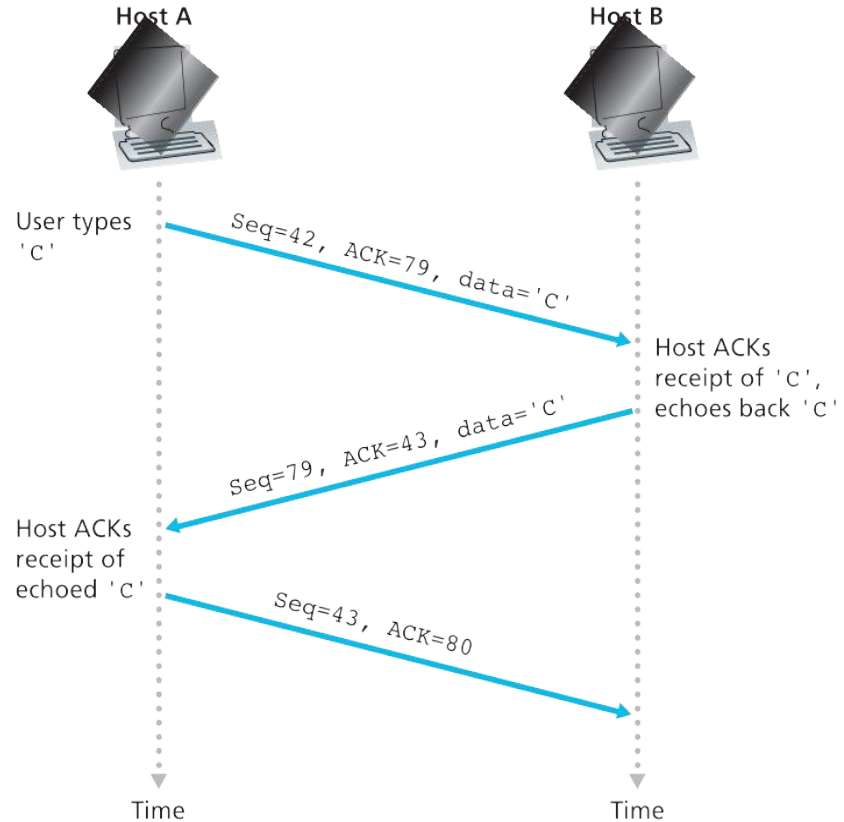




- Header Length
 - (in 32 bit words)
- ACK
 - ACK is an acknowledgement
- RST, SYN, FIN
 - connection setup and teardown
- URG, PSH
 - normally not used, pass to upper layer
- Options can be used to negotiate
 - e.g. maximum segment size (MSS)
- Receive Window
 - for flow control



- Counts bytes in the data stream
 - not segments
- Sequence number of a segment is (42)
 - first byte in the data field
- ACK (43) states that all bytes up to byte 42 has been received and is now waiting for bytes 43 onward.



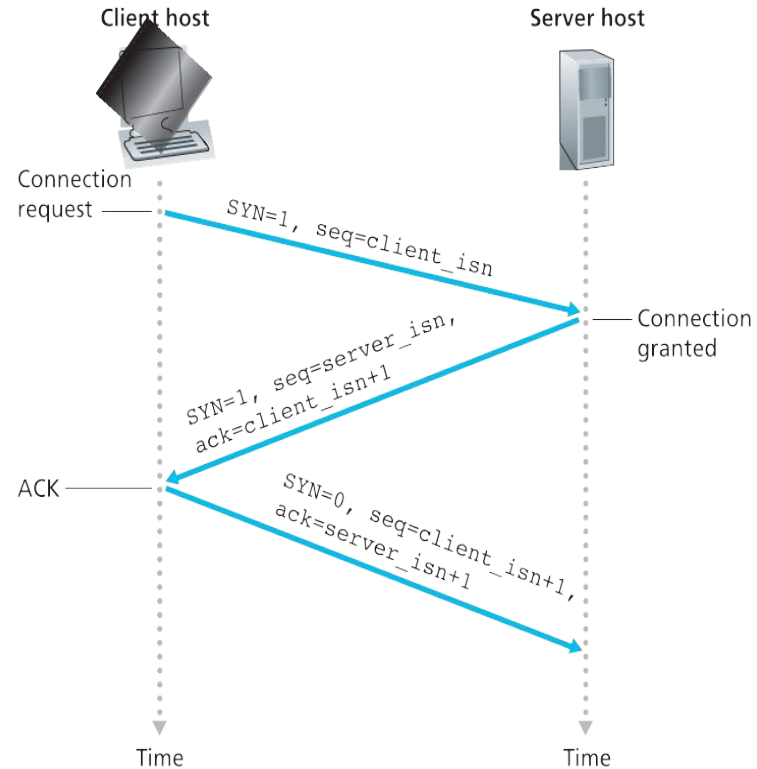


- Three Way Handshake
- SYN segment

Does not contain application data. Chooses random sequence number.

SYN/ACK

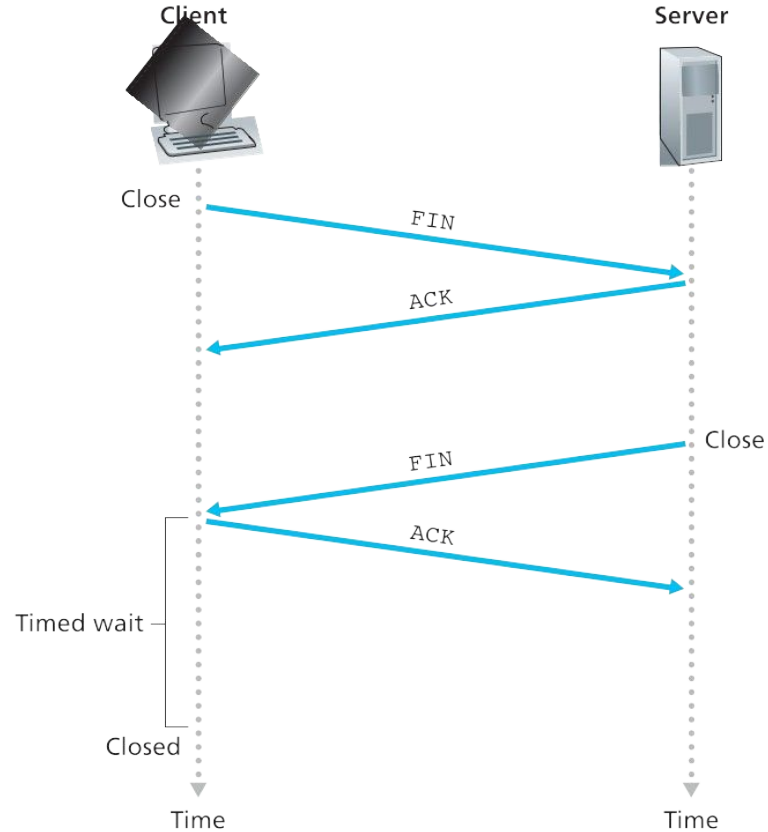
- Does not contain application data, chooses random sequence number
- SYN=0 (Connection is established)
May contain application data



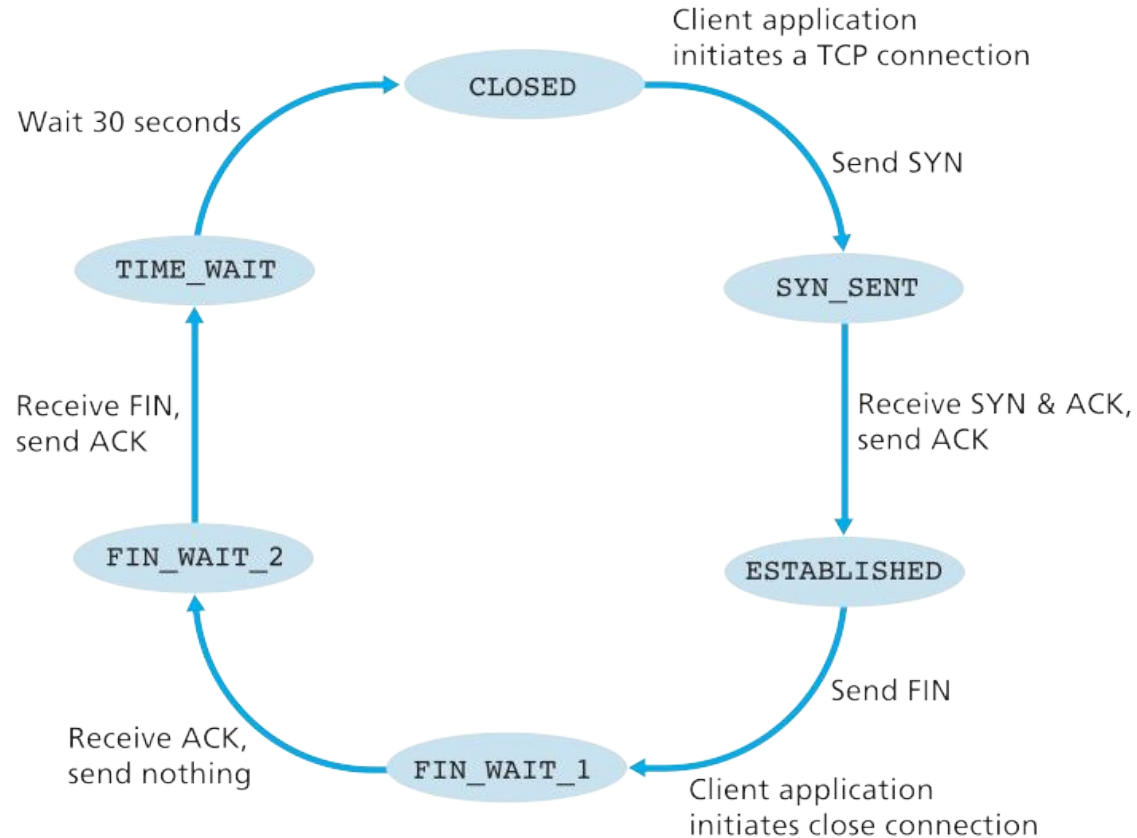


TCP Connection Tear Down

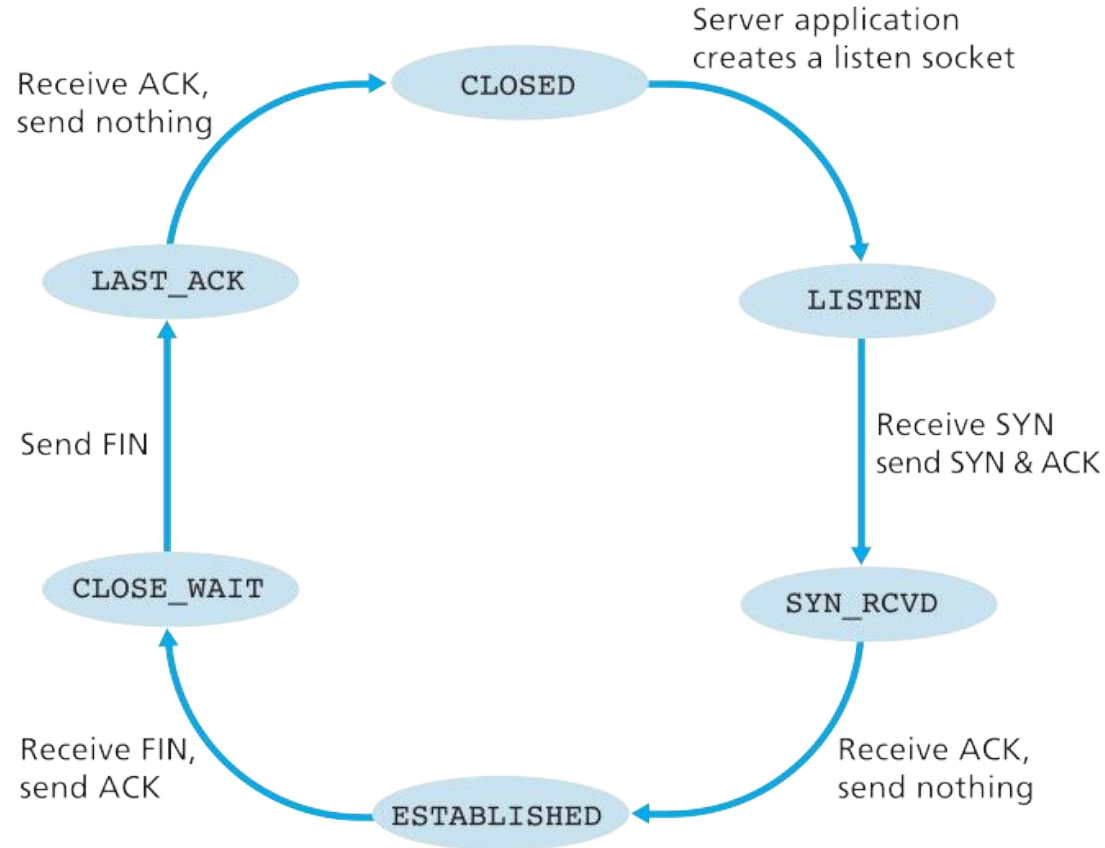
- FIN initiates the connection shutdown
- They are mutual, and acknowledged
- Allows both parties to free allocated resources



TCP Typical State Transitions

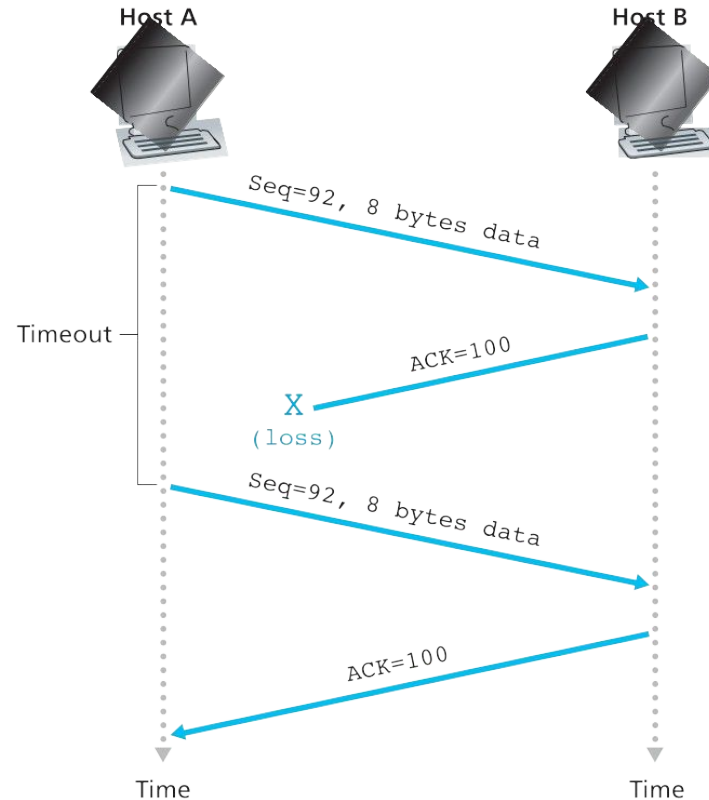


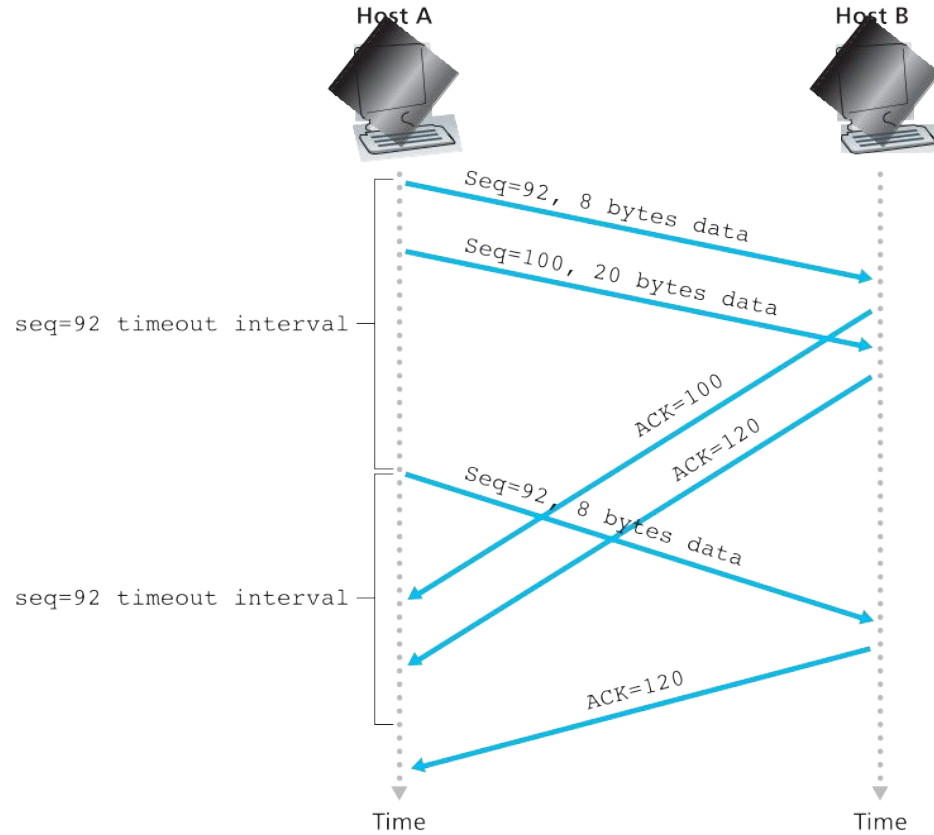
TCP Typical State Transitions

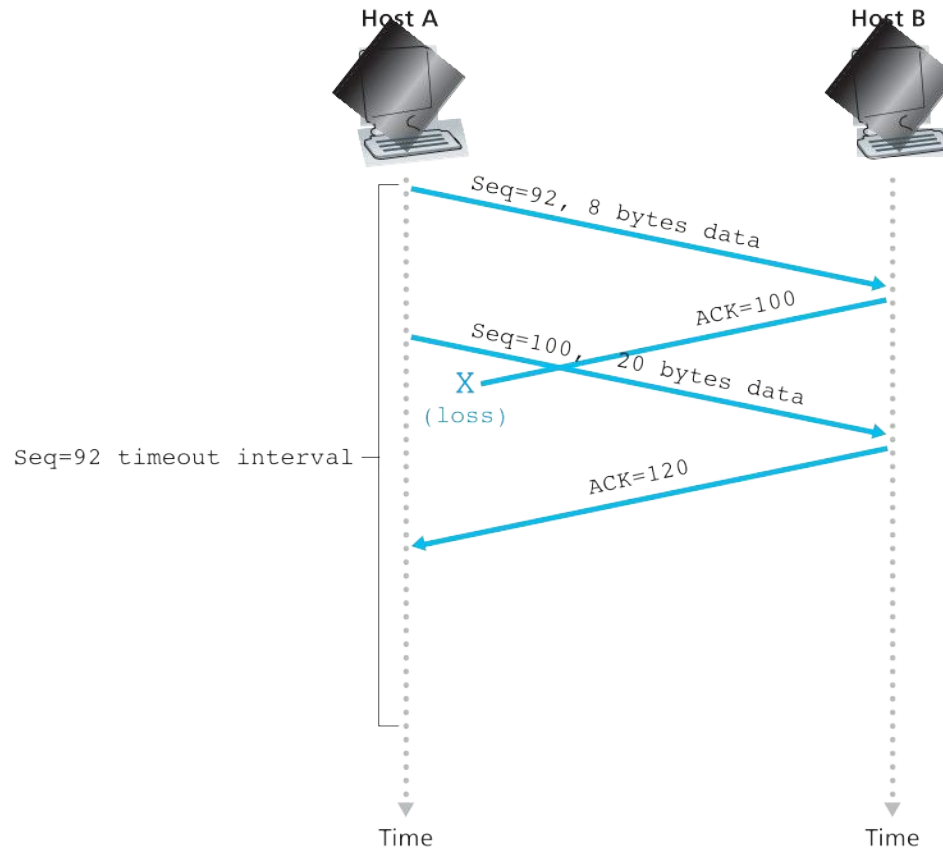




- Retransmission due to lost ACK
- A timeouts on segment 92 and resends
- B realises that segment 92 has already been received and discards
- Retransmits ACK again

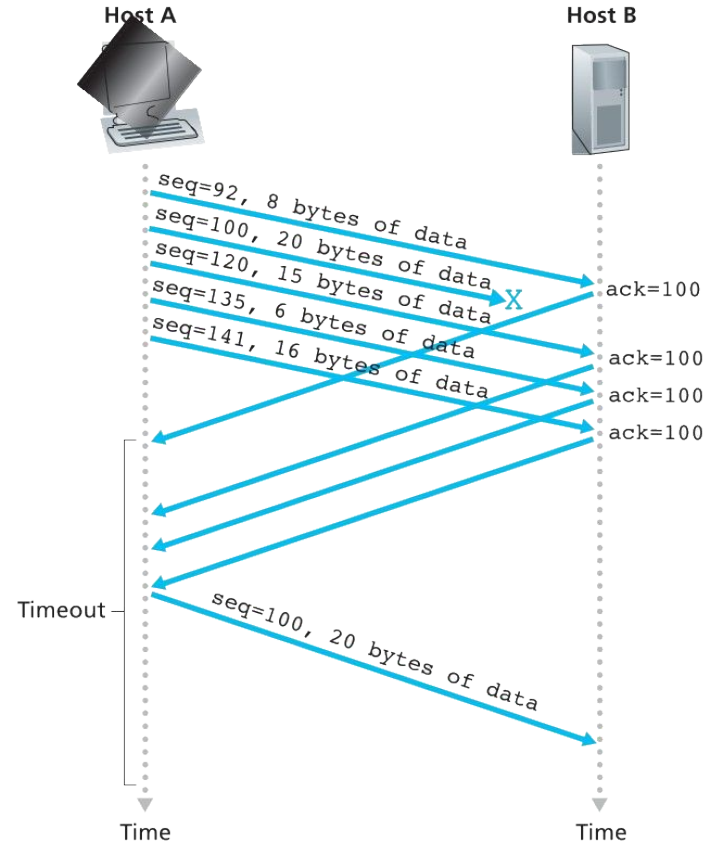






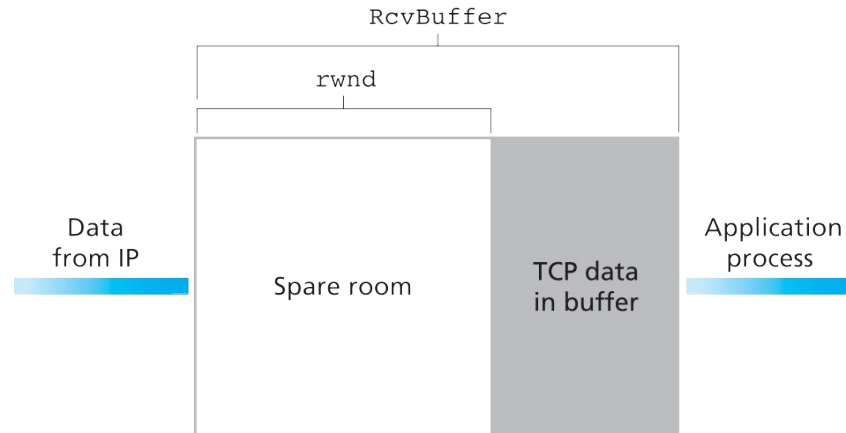


- Does not wait for timeout





- TCP provides a flow-control service
- Speed matching service, avoid overflowing receiver's buffer
- Sender maintains receive window (rwnd), telling it how much buffer space is available at the receiver. Sender keeps: $\text{LastByteSent} - \text{LastByteAcked} \leq \text{rwnd}$





- Remember that the server allocates resources for every TCP SYN?
- This can be used in a DoS attack by exhausting the server's connections
 - leaving them open by not completing the third part of the handshake

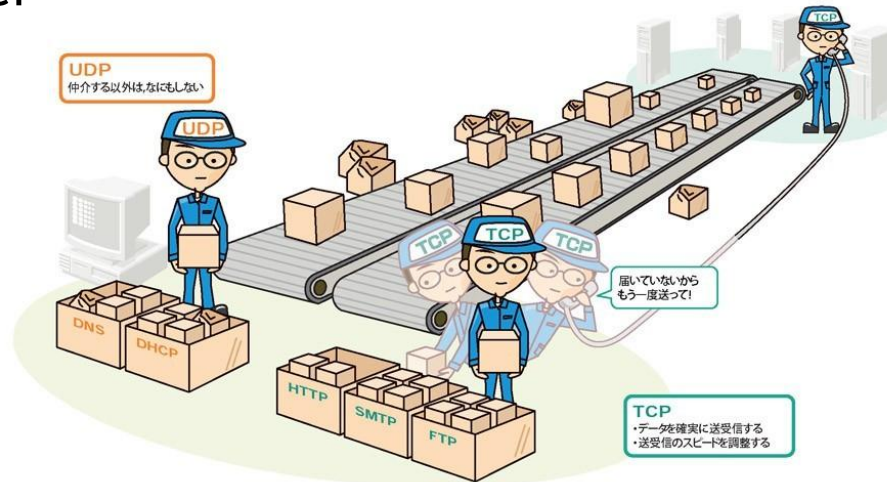
A server needs to await a time-out before releasing the resource



- Mitigation for this are SYN cookies:
 - Server crafts special sequence number, without allocating resources for its SYNACK
 - When it receives the ACK from a client it verifies that this corresponds to an earlier SYN (SYNACK) and established the connection
 - If no ACK is send, no resources are ever allocated



- Principles of data transfer
- Transport Layer services
- Ping/ICMP
- UDP
- TCP





Thank You!

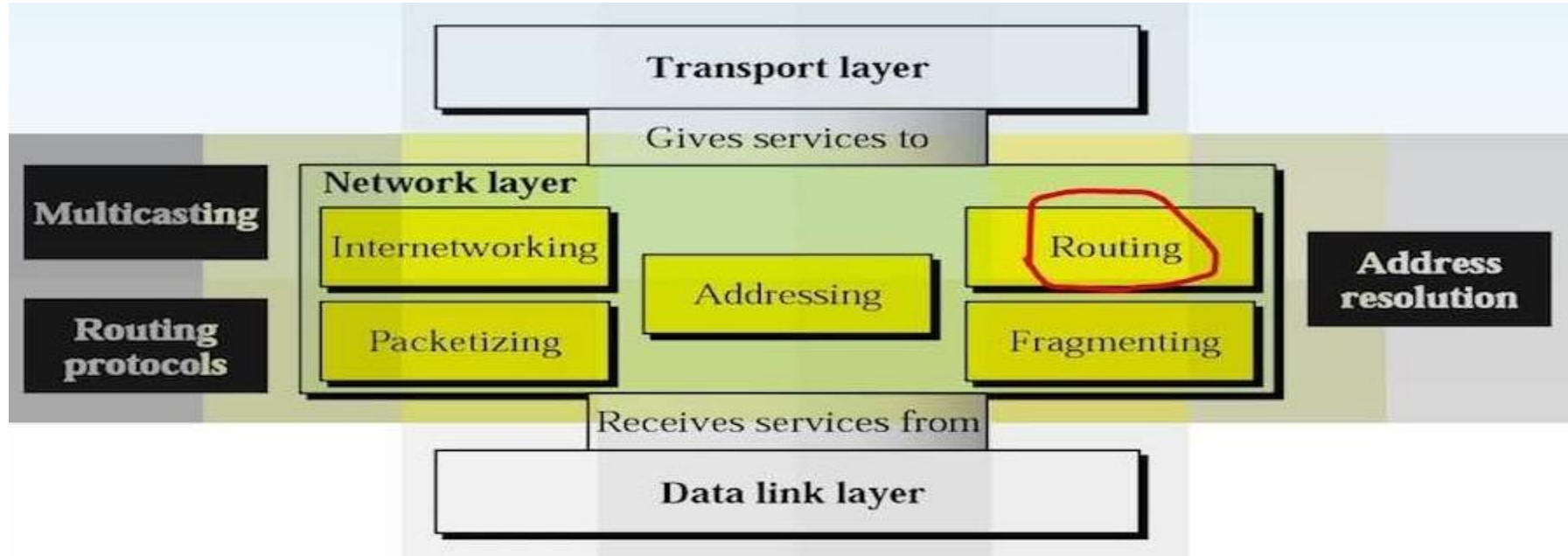


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*The Network
Layer*



Lecture Content

- . Network Layer services
- . Datagram networks
- IP Addresses
- IP protocol
- Routers: forwarding



References

- Kurose & Ross; Computer Networking; Chapter 4 & 5
- http://www.tcpipguide.com/free/t_NetworkLayerLayer3.htm
- <https://www.geeksforgeeks.org/differences-between-virtual-circuits-datagram-networks/>
- <https://www.lifewire.com/top-network-routing-protocols-explained-81796>
- https://www.tutorialspoint.com/network_security/network_security_data_link_layer.htm
- https://en.wikipedia.org/wiki/Network_address_translation
- <http://www.networkerinterview.net/entries/nat/nat-network-address-translation->



Post lecture discussion topics

- Read the reference about NATting
- Research and compare two network layer routing protocols
- List and describe the purpose of each field within the headers of a link-layer frame compared to a datagram
- packet

What are the known security issues of the network layer?

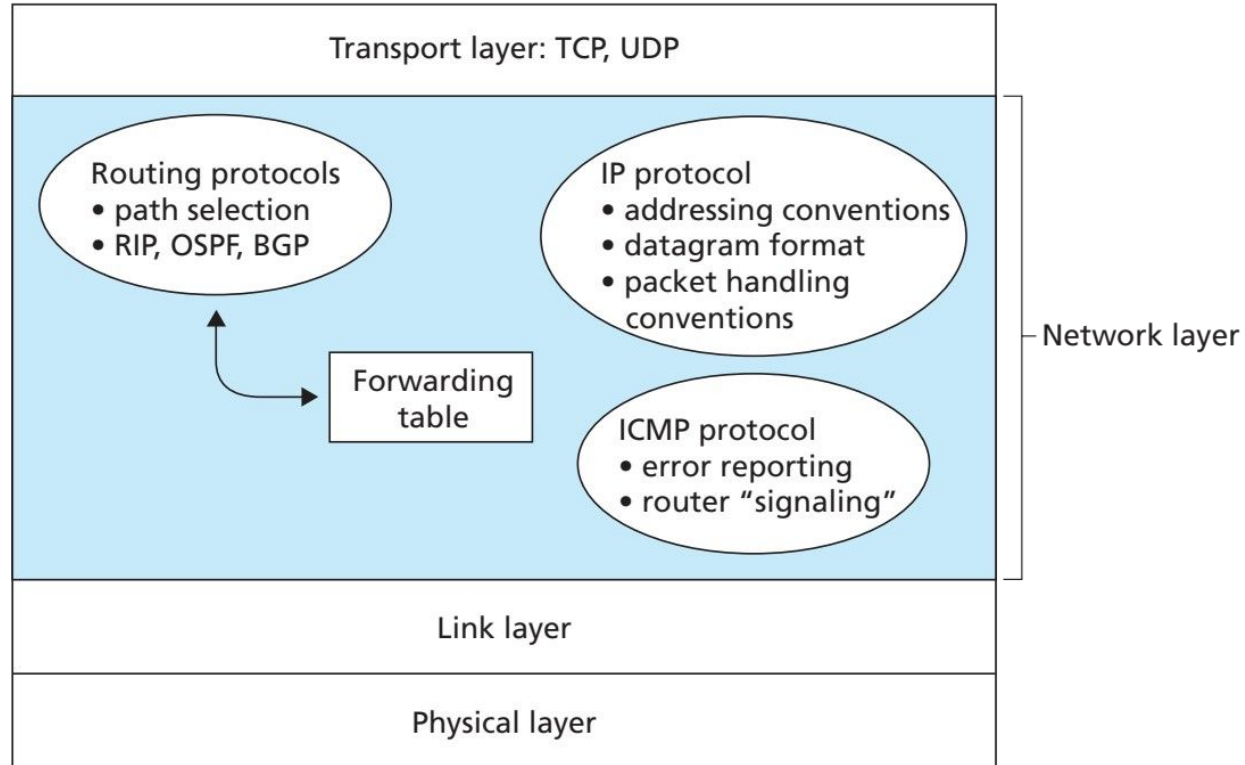


Lecture Content

- **Network Layer**
- **services**
- Datagram networks
- IP Addresses
- IP protocol
- Routers: forwarding



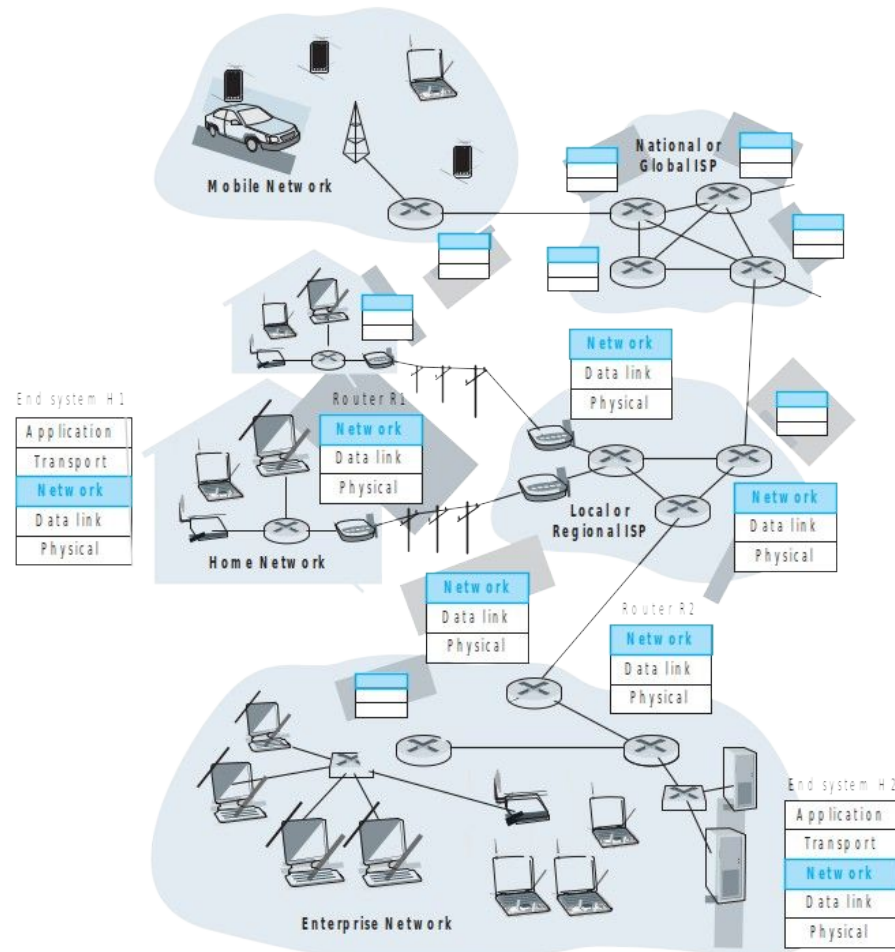
Inside the Network Layer





Functions of the Network Layer

- Data Plane
 - local logic
- Control Plane
 - network-wide logic
- Forwarding/Routing
 - determine the path taken by packets through the network
 - network-wide activity
 - determines and adjust forwarding tables





Services the Network Layer can provide

- Guaranteed Delivery:
 - packet will eventually arrive
- Guaranteed Delivery with bounded delay:
 - packet will arrive within a certain time
- In-order packet delivery:
 - packets arrive at destination in order they were sent



Services the Network Layer can provide

- Guaranteed minimal bandwidth:
 - emulates a specific bit rate (e.g. 1 Mbps)
- Guaranteed maximal jitter:
 - time between the arrival of two packets is equal to the time between sending the packets
 - with a defined margin of error
- Security Services:
 - integrity
 - confidentiality
 - authentication



Lecture Content

- . Network Layer services
- . **Datagram networks**
 - IP Addresses
 - IP protocol
 - Routers: forwarding

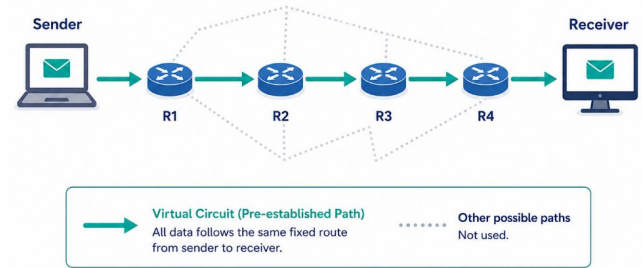


Network Service Types

- Virtual Circuits
 - connection-oriented service

Virtual Circuits

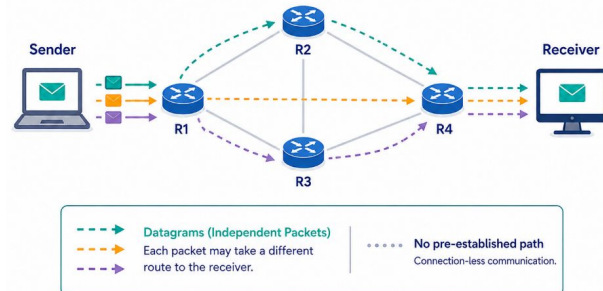
connection-oriented service



Datagram networks

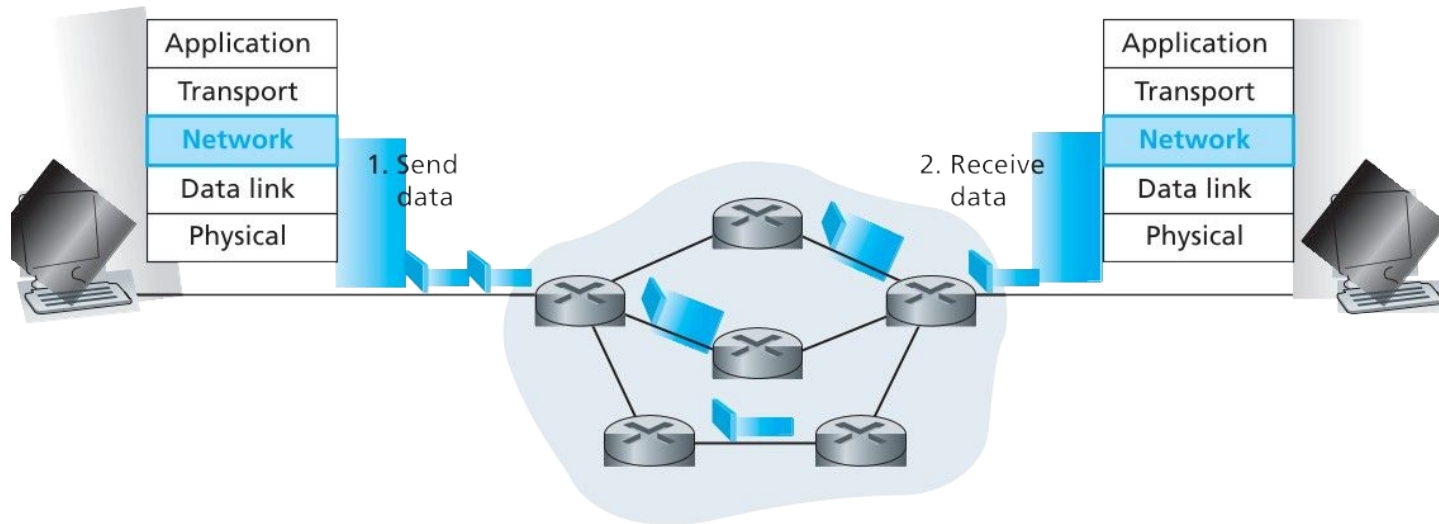
connection-less services

- Datagram networks
 - connection-less services



Datagram Network

- Each datagram (packet) is forwarded independently based upon its destination address
 - in the header field





Datagram Network

- Router forwards packets based on a *forwarding table*
 - maps from destination addresses to interfaces
- Matching is performed on a longest prefix basis

Prefix Match	Link Interface
11001000 00010111 00010	0
11001000 00010111 00011000	1
11001000 00010111 00011	2
otherwise	3

Lecture Content

- . Network Layer services
- . Datagram networks
- **IP Addresses**
- IP protocol
- Routers: forwarding



Internet Protocol Addresses

- IP address
 - numerical label assigned to each device connected to a computer network that uses the Internet Protocol for communication
- Two main functions
 - host number
 - NIC identification
 - (network interface card)



Internet Protocol Addresses

- IP address space managed
 - Globally
 - Internet Assigned Numbers Authority (IANA)
 - Regionally
 - 5 Regional Internet registries (RIRs)
 - responsible in their designated territories for assignment to local Internet registries
 - eg ISPs, and other end users

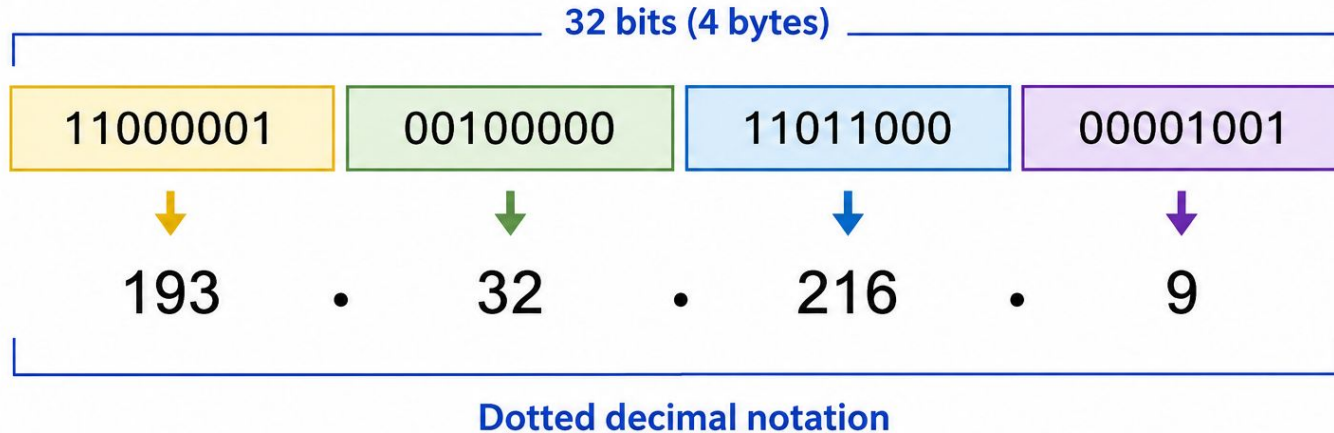
Internet Protocol Addresses

- IPv4 addresses were distributed by IANA to the RIRs in blocks of approximately 16.8 million addresses each
 - exhausted at the IANA level since 2011 some IPv4 addresses are reserved for private networks and are not globally unique
- DMU uses the class A 10 network
 - look at the IP address on your phone ...



IP Addresses

- IPv4 addresses are **32 bits** long (4 bytes)
- There are 2^{32} = approx. **4 billion** possible addresses
- Typically written in **dotted decimal notation** (octets):



IP Addresses

- Originally, IPv4 address space was split up into 5 “classes”
 - A to E
- Talk about “/8” (pronounced “slash 8”) blocks
 - 256 such blocks
 - numbered 0/8 to 255/8
 - Each /8 block has 254 (about 16.7 million) addresses

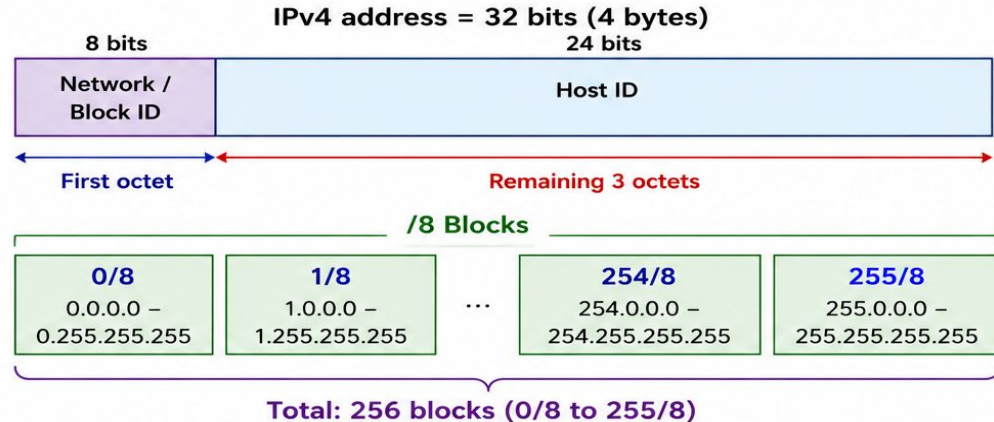
What is a /8 block?

A /8 block means the first 8 bits (1 octet) are the network/block identifier.

The remaining 24 bits are for hosts in that block.

Addresses in each /8 block

- Total addresses per block = 2^{24}
= 16,777,216
- Usable host addresses = $2^{24} - 2$
= 16,777,214 (≈ 16.7 million)



Using /8 blocks helped in early IP address allocation and routing summarization, making routing tables smaller and more efficient.



Subnet Masking

- IP address classes: C =/24 ; B =/16 ; A =/8
- Subnet mask “masks” the subnet bits
- ie
 - subnet mask = 255.255.255.0/24
 - class C network
 - subnet mask = 255.255.0.0/16
 - class B network
- Eg IPv4 address = 192.0.2.1
 - subnet mask =
255.255.255.0
 - 254 possible host addresses
 - but one address is the network gateway



Local loopback

IPv4 loopback address

127.0.0.1

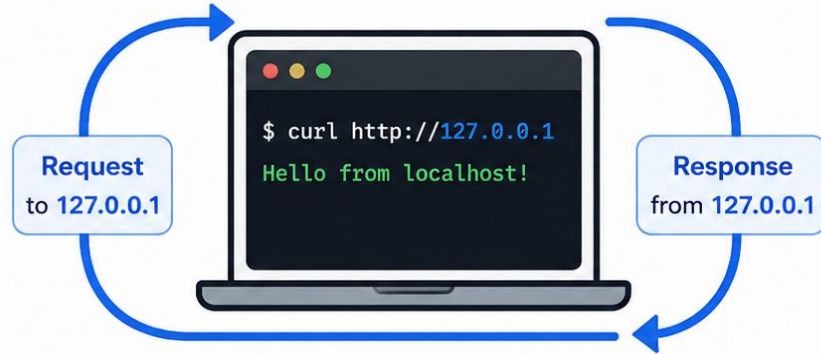
Hostname

localhost



Used for testing

Traffic sent to the loopback address
stays on the same machine.



All traffic stays on this machine.
It **does not** leave the system.



What is it?

A special IP address that refers to this computer.



Why use it?

Test services and applications without a network.



Safe & local

Perfect for development, testing, and troubleshooting.

IP Addresses

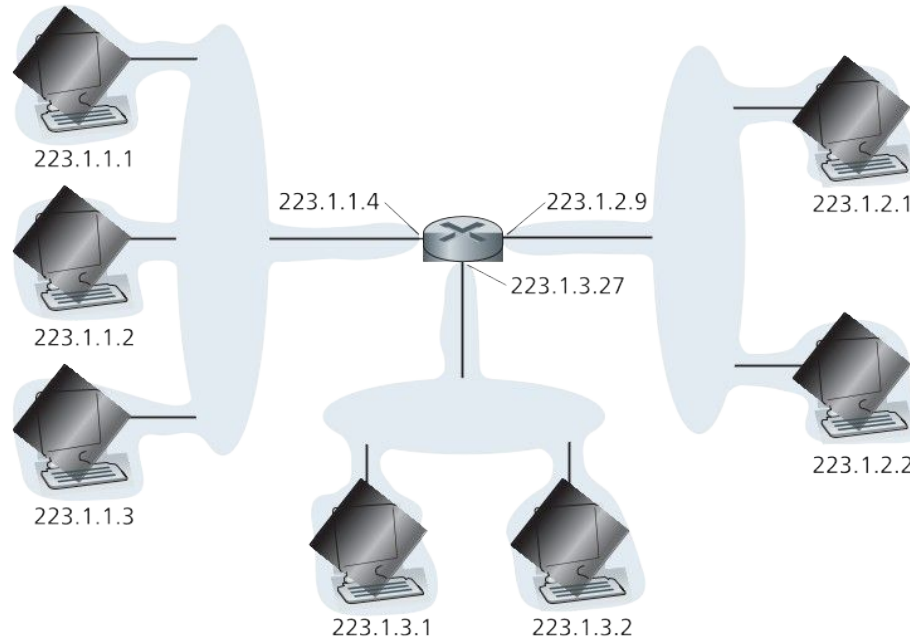
Three class C local subnets connected by a
router netmask = 255.255.255.0

4th octet = host machine number

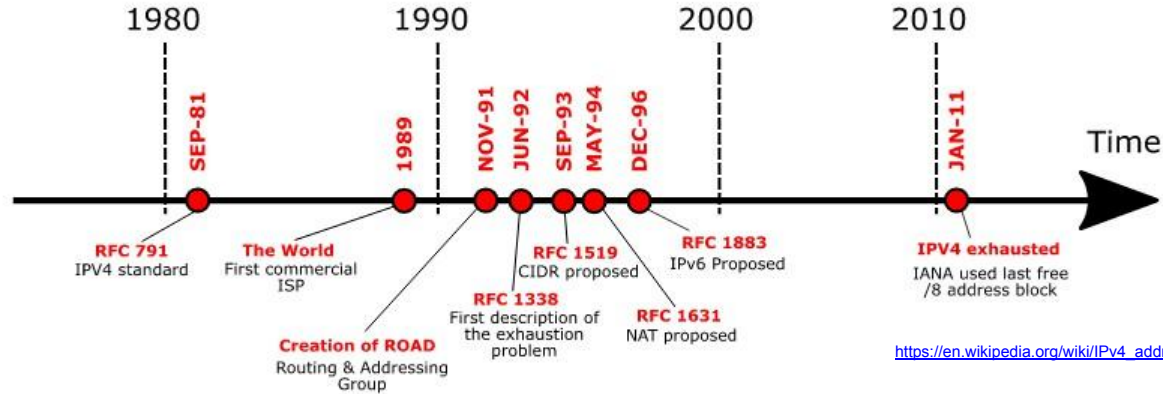
223.1.1.0/24

223.1.2.0/24

223.1.3.0/24



IP Address Exhaustion



- IPv4 address space was exhausted in 2011
 - anticipated in ~1988
- Solutions
 - CIDR
 - Network Address Translation
 - IPv6 (128 bit addresses) - 1998



CIDR

- Classless Interdomain Routing
 - (CIDR, RFC 4632)
- The 32 bit IP address is divided into two parts
 - written as a.b.c.d/x
 - where x is the number of bits in the first part of the address
- The x most significant bits are known as the network prefix
- Router only needs to consider network prefix



- Variable length subnet masks

200.23.16.0/26

11001000 00010111 00010000 00000000

$2^6 - 2 = 62$ host addresses

- 200.23.16.0/20

11001000 00010111 00010000

00000000 $2^{12} - 2 = 4,094$ host addresses

- 199.31.0.0/16

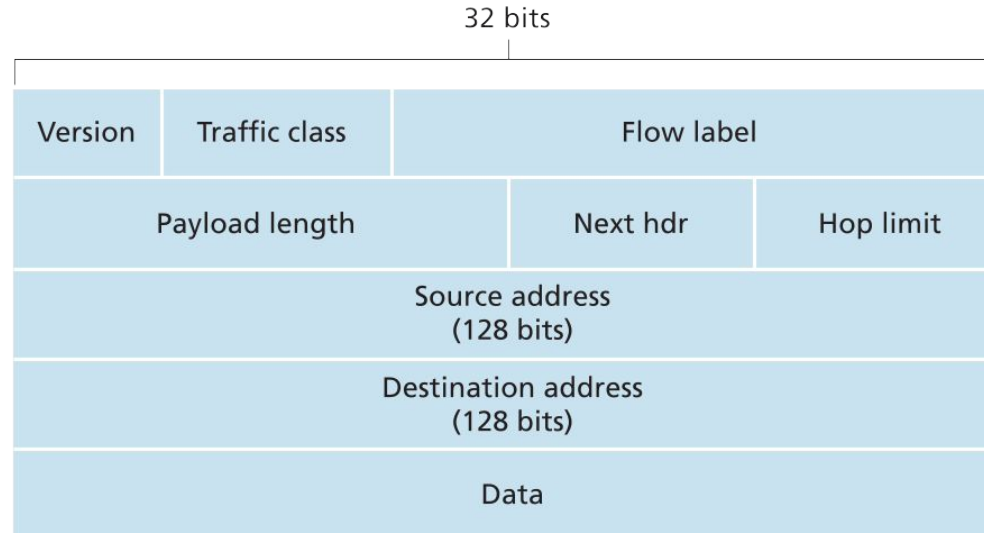
11000111 00011111 00000000 00000000

$2^{16} - 2 = 65,534$ host addresses

- [Try this CIDR subnet calculator](#)

IPv6

- IPv6 increased address space to 128 bits
- Streamlined 40 byte fixed-length header
- Traffic Class can be used to give priorities.
- Flow Label is used to mark packets to belong to a particular flow, e.g. for special QoS requirements or real-time traffic
- RFC1752 and 2460





IPv6 vs IPv4

- IPv6 does not allow fragmentation
 - if a packet is too large to transmit, the router returns an ICMP error message
 - RFC 4443
ICMPv6
 - sending host must resubmit smaller datagrams
 - MTU is indicated in the ICMP error message
- IPv6 does not do a header check sum
 - this is already done at link-layer and transport-layer
 - saves processing time

How to get an IP address

- How does a host get an IP address in the first place?
- Two options:
 - Static IP address assigned by an administrator
 - Dynamic IP address obtained through Dynamic Host Configuration Protocol
 - service that “leases” IP addresses to hosts

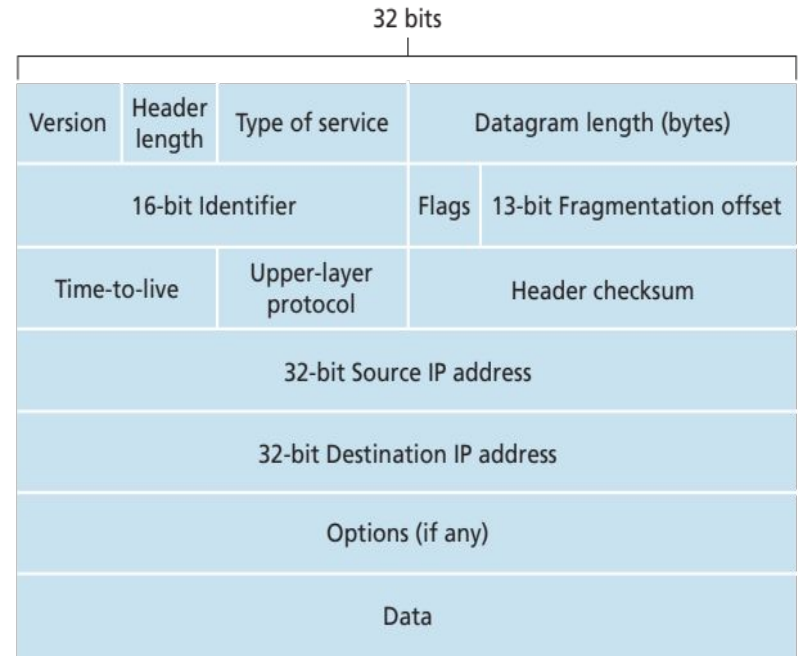
Lecture Content

- . Network Layer services
- . Datagram networks
- IP Addresses
- **IP protocol**
- Routers: forwarding



IPv4 Datagrams

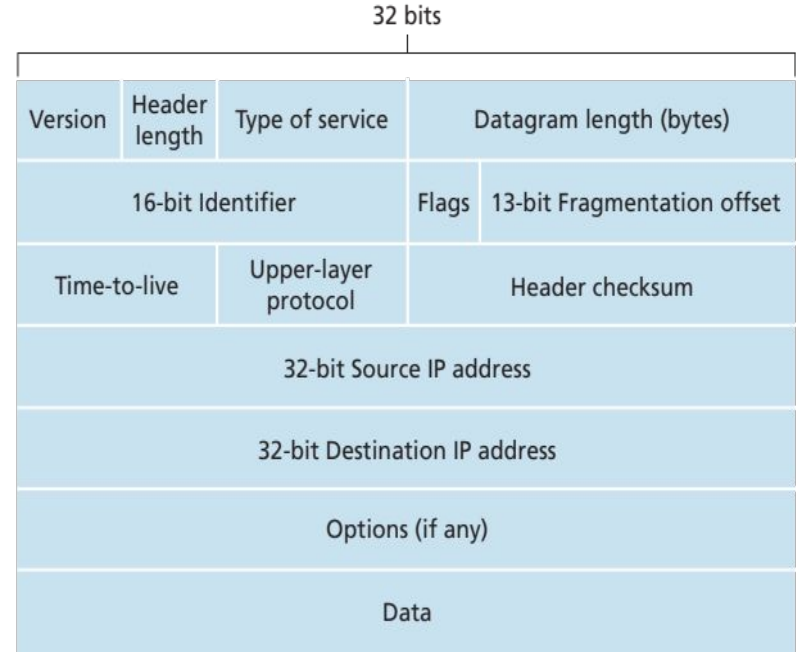
- Version number
- Header length (20 bytes w/o options)
- Type of service to distinguish different uses such as for real-time datagrams
- Datagram length (total, in bytes), max 65,535 bytes ($= 2^{16} - 1$)





IPv4 Datagrams

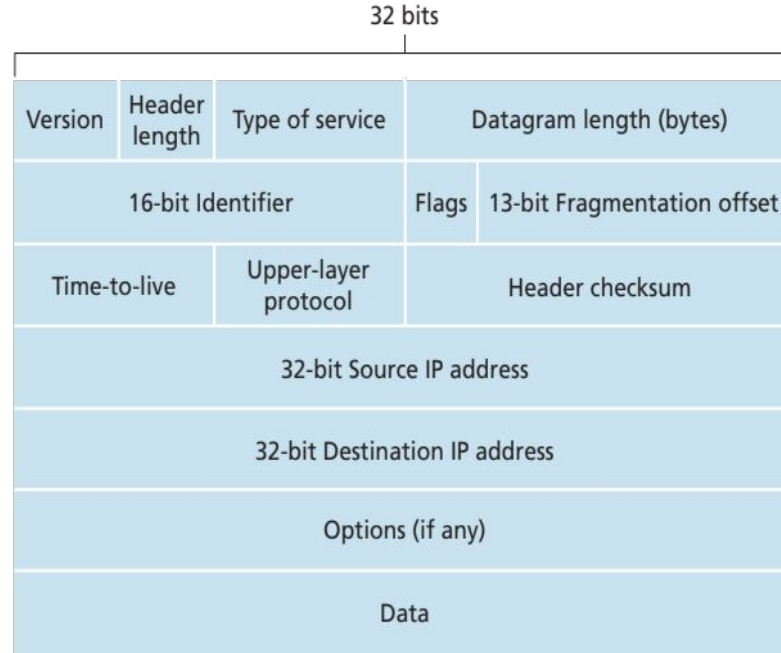
- Identifier, flags, fragmentation offset
- time-to-live (TTL)
- ensures datagrams do not circulate forever
- decremented at each router





IPv4 Datagrams

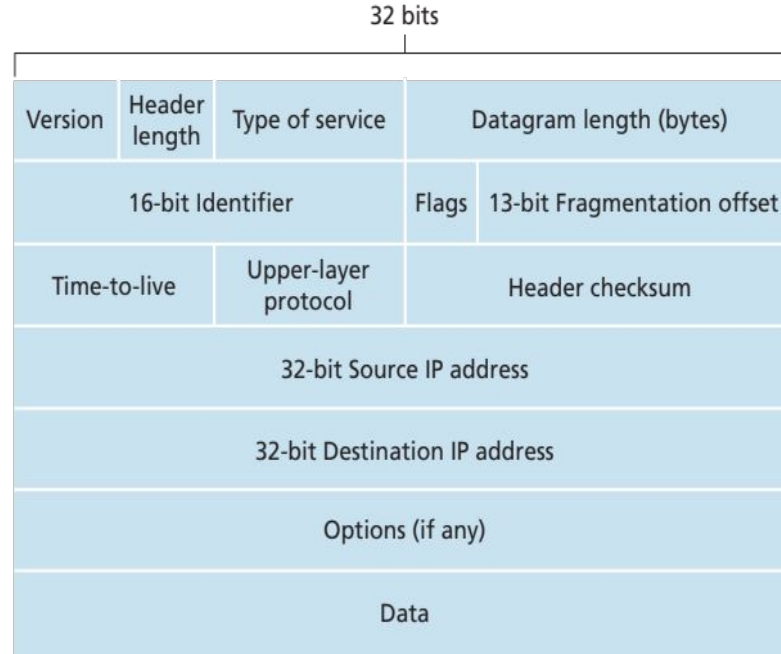
- Protocol:
- Glue to the transport layer
 - TCP=6, UDP=17
- Header checksum
- Needs to be recomputed at every router
- Normal to discard datagrams that are corrupted





IPv4 Datagrams

- Source and Destination IP addresses
- Options (used rarely and not part of IPv6)
- Data
- Normally contains a transport layer segment (TCP/UDP) or ICMP message





Lecture Content

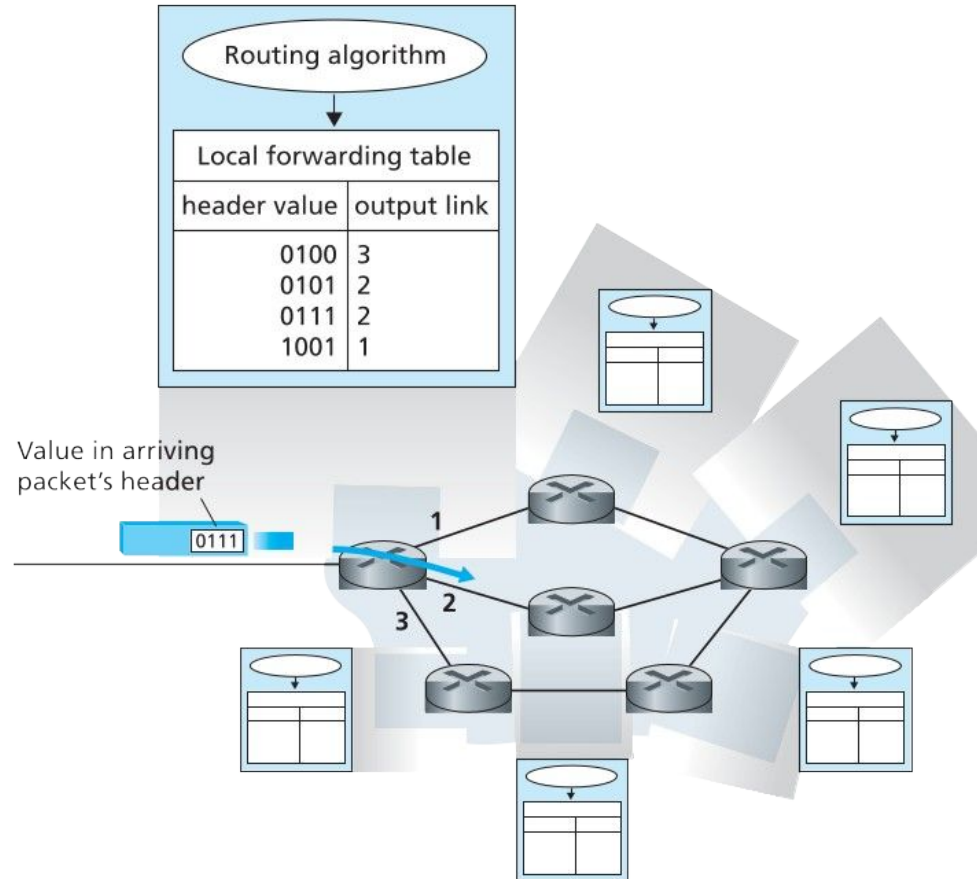
- . Network Layer services
- . Datagram networks
- IP Addresses
- IP protocol
- **Routers: forwarding**



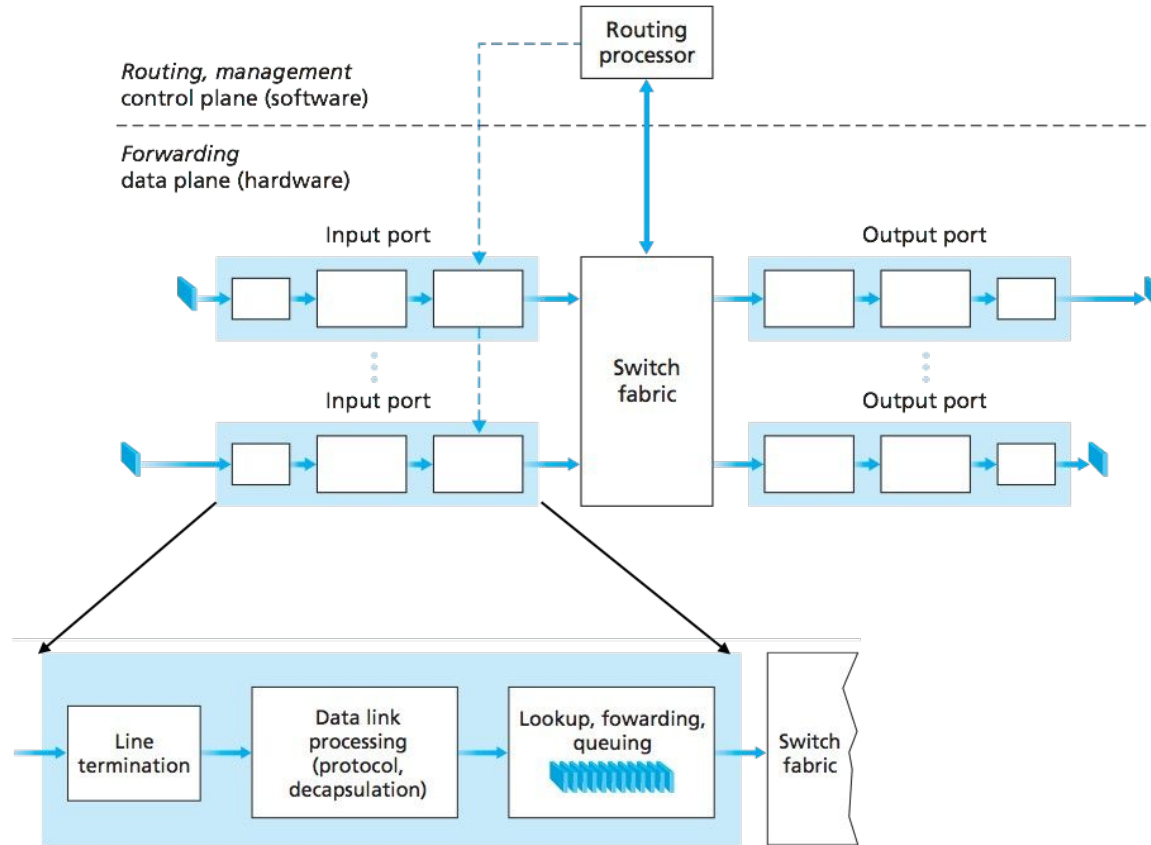
What is a Router?

- Device that connects two or more packet-switched networks or subnetworks
- Two primary functions:
 - manages traffic between these networks by forwarding data packets to their intended IP addresses
 - allows multiple devices to use the same Internet connection
- Routing decision made by reference to a routing table

Example of forwarding tables



Inside a Router





Routing Protocols

- **Routing Information Protocol**
- Open Shortest Path First
- Border Gateway Protocol



Routing Protocols

- . Routing Information Protocol
- . **Open Shortest Path First**
- . Border Gateway Protocol

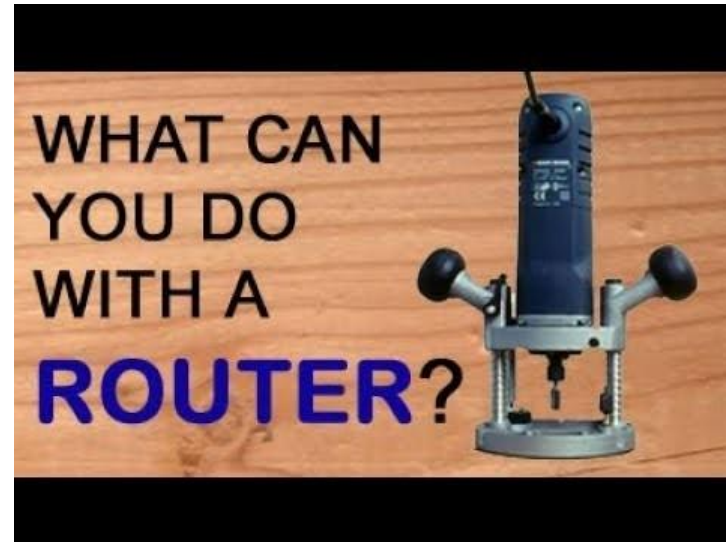


Routing Protocols

- . Routing Information Protocol
- . Open Shortest Path First
- . **Border Gateway Protocol**

Lecture Contents

- Network Layer services
- Datagram networks
- IP Addresses
- IP protocol
- Routers:
 - forwarding





Thank You!



Operating Systems & Networks

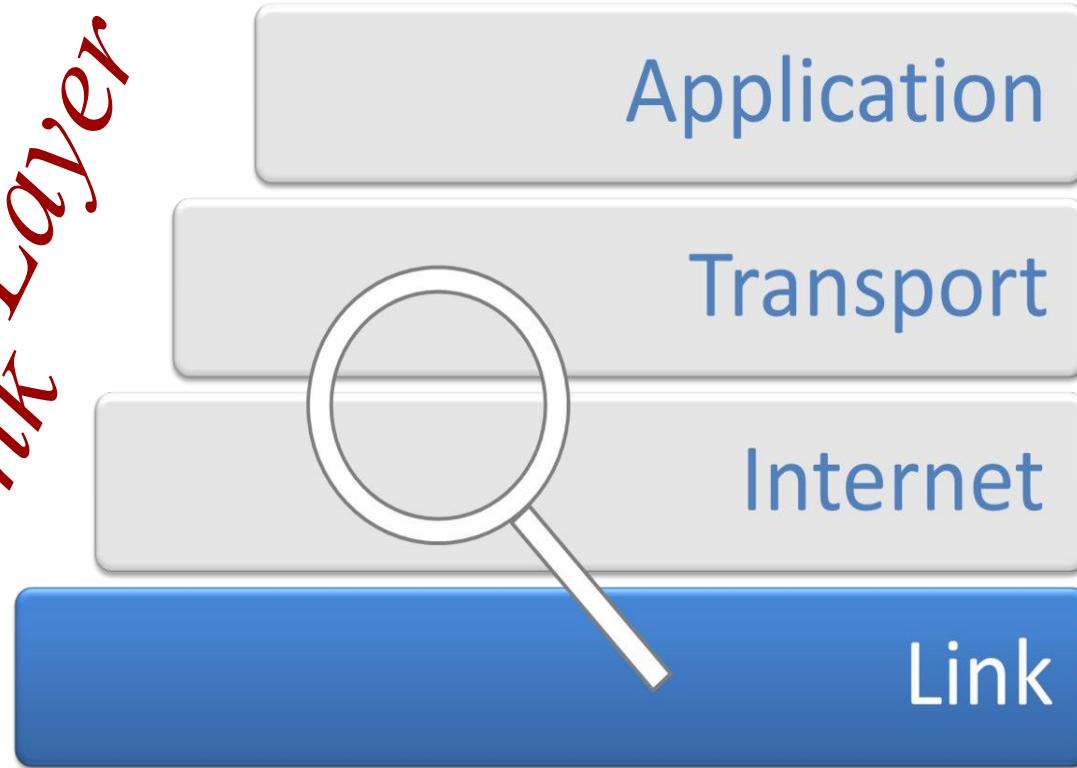
CTEC1704C

Lecture – 13 : The Link Layer

Dr. Sameer Jha, SFHEA
Faculty of Computing
De Montfort University Cambodia



The Link Layer



CTEC1704 -

Lecture Content

- Principles of Link Layer services
- Link Layer protocols
- Link Layer Addressing
- Ethernet Protocol
- Switches and Hubs



- Kurose & Ross; Computer Networking; Chapter 6
- <https://www.tutorialspoint.com/error-detection-and-correction-in-data-link-layer>
- https://en.wikipedia.org/wiki/Network_switch
- https://www.tutorialspoint.com/network_security/network_security_data_link_layer.htm
- <https://en.wikipedia.org/wiki/Ethernet>
- <https://www.electronicdesign.com/technologies/communications/article/21801558/3-new-versions-of-ethernet-you-should-know-about>
- https://en.wikipedia.org/wiki/Network_switch
- https://www.tutorialspoint.com/network_security/network_security_data_link_layer.htm

Post lecture discussion topics

- What are the contents of the arp table of the (Linux) computer in front of you?
- Research, name and describe three error detection strategies used by the link layer
- What are the known security issues of the link
- layer? Compare and contrast 10BASE-T with
- NBASE-T What is the current standard?

Explain the role and functionality of the Spanning Tree Protocol

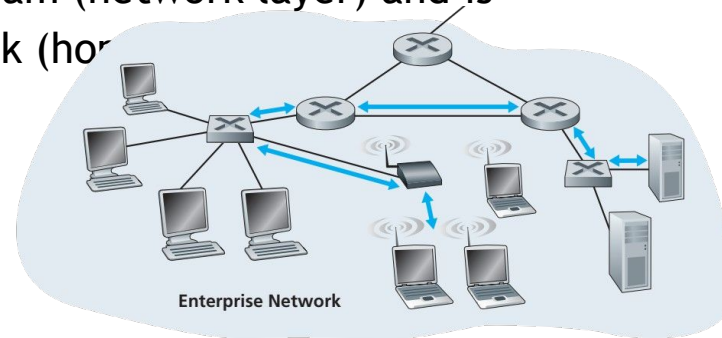
Lecture Content

- **Principles of Link Layer services**
- Link Layer protocols
- Link Layer Addressing
- Ethernet Protocol
- Switches and Hubs



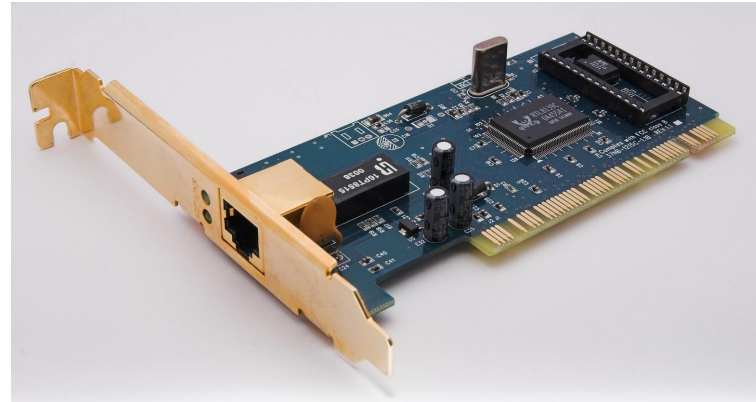
Nodes, Links and hops

- Node -
 - device that runs a link-layer protocol
 - eg hosts, routers, switches, hubs, etc
- Link -
 - comms channel that connects two adjacent nodes
- Link-layer *frame*
 - encapsulates the datagram (network layer) and is transmitted over the link (hop)



Nodes, Links and hops

- Link-layer
 - typically implemented in the Network Adapter or Network Interface Card (NIC)



Link Layer Services

- Frames
 - Link-layer protocols encapsulate datagrams from the network layer before transmission
 - Frame header added to each datagram



Link Layer Services

- Link Access
 - Medium Access Protocols (MAC) specifies how a frame is transmitted onto the link
 - Simple for point to point links
 - More sophisticated when multiple access is required



Link Layer Services

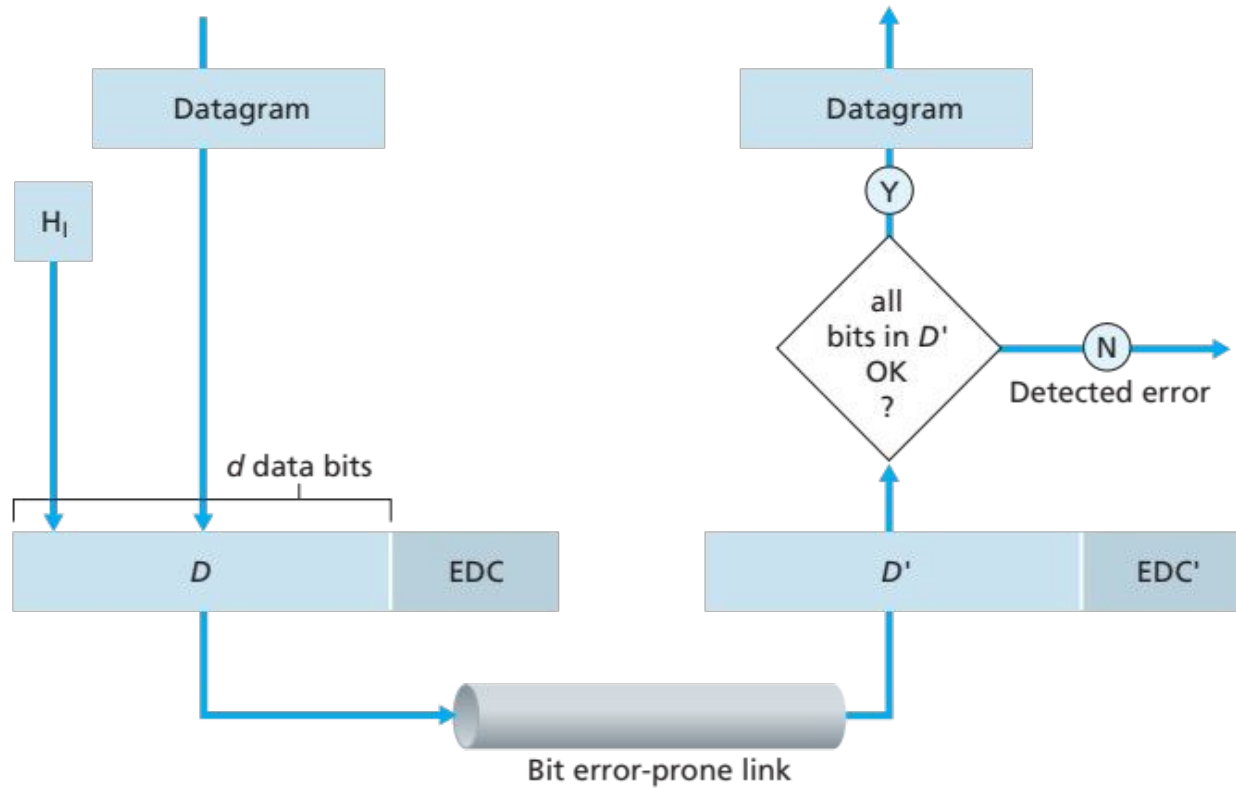
- Reliable Delivery
 - Guarantees delivery of each network-layer datagram across the link
 - Typically used when there are high error-rates
 - otherwise, TCP protocol (transport layer) serves a similar function and overheads can be avoided



Link Layer Services

- Error Detection and Correction
 - can detect flipped bits in a frame
 - integrity check
 - typically implemented in hardware and also able to correct errors

Link Layer Services



Error detection

- Types of errors:
 - **Single** bit error
 - In the received frame:
 - only one bit has been flipped (corrupted)
 - **Multiple** bits error
 - more than one bits are flipped
 - **Burst** error
 - more than one consecutive bits are flipped



Error Detection

- Parity Bits
 - parity bit can detect bit errors
- Cyclic redundancy check
 - CRC
 - similar in concept to the checksum in transport layer
 - more powerful error-detection coding: can detect multiple bit errors

Lecture Content

- Principles of Link Layer services
- **Link Layer protocols**
- Link Layer Addressing
- Ethernet Protocol
- Switches and Hubs

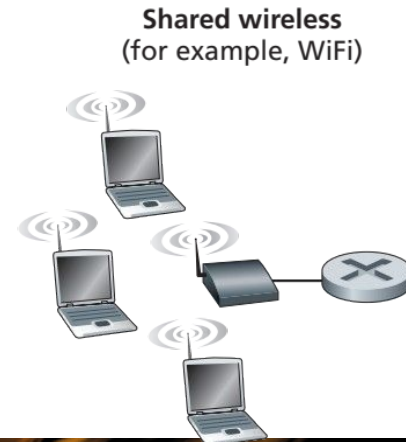
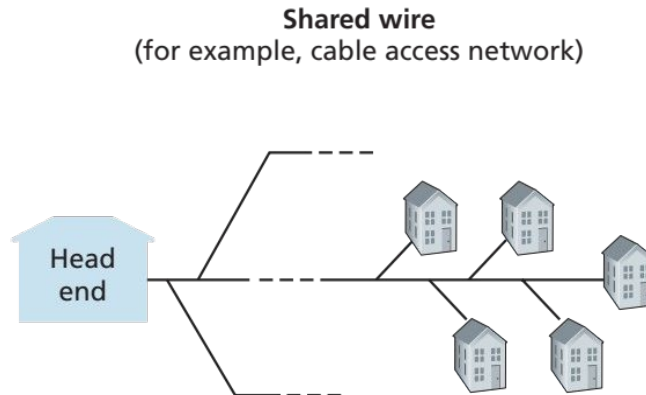


Link Access

- Broadcast Link

Multiple sending and receiving nodes connected to a single broadcast channel.
Examples are Ethernet and Wireless LANs

- Problem: How do we make sure the link is used fairly?
- Multiple Access Protocols detect and avoid collision of transmitted frames
- Distinguish Channel Partitioning, Random Access and Taking Turns approaches.





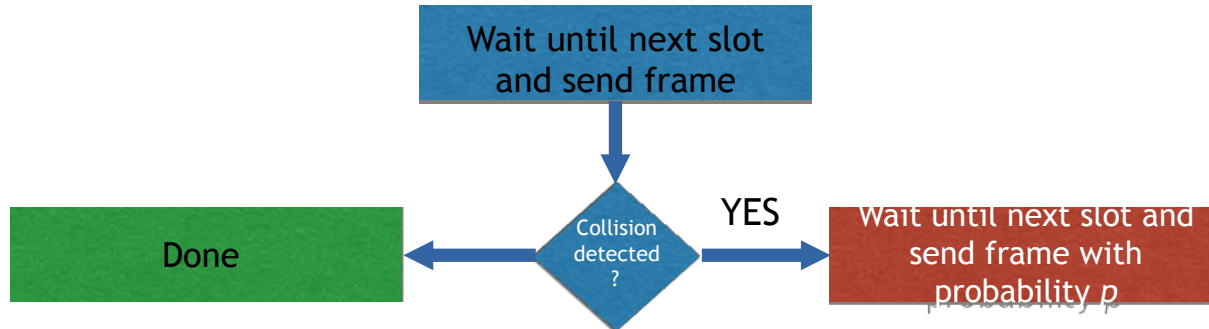
Link Access

- Time Division Multiplexing
 - divide time into time frames divide
 - each time frame into slots
 - each slot is allocated to a specific node: that can send during its allocated slot only
- Frequency Division Multiplexing
 - divides the channel into different frequencies each
 - node is assigned a frequency
- Code Division Multiple Access
 - each node is given a separate code
 - nodes can submit simultaneously



Random Access

- Slotted ALOHA
 - Frames are equal size
 - Time is divided into equal sized slots (= enough to transmit one frame)
 - Slots are synchronised and nodes transmit only in the beginning of slots
 - When frames collide all nodes detect collision before the slots ends





Carrier Sense Multiple Access (CSMA - CD)

1. Adapter prepares a link-layer frame, and puts the frame in the adapter buffer.
 - 2a. If channel is idle it starts to transmit the frame.
 - 2b. If channel is busy it waits until it senses no signal energy and then starts to transmit
 3. During transmission adapter monitors signals to detect collisions.
 - 4a. If adapter transmits the frame without detecting a collision, the frame is sent.
 - 4b. If adapter detects a collision it aborts the transmission and sends a jamming signal.
 5. After aborting, the adapter waits a random time and returns to step 2
- NB worst case scenario - the packet is never successfully sent!

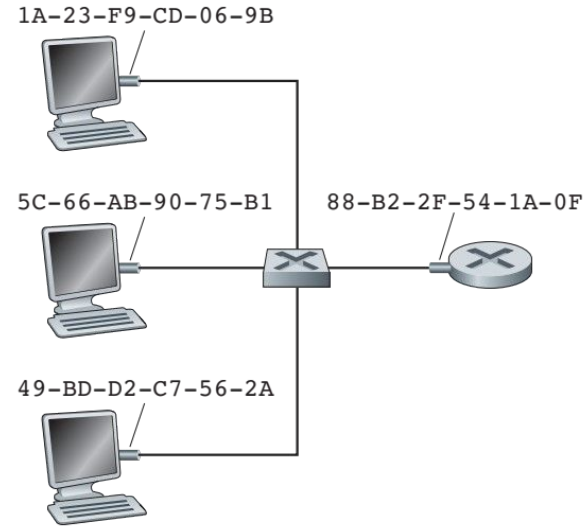
Lecture Content

- Principles of Link Layer services
- Link Layer protocols
- **Link Layer Addressing**
- Ethernet Protocol
- Switches and Hubs



Link Layer Addresses

- **MAC address** (also called LAN address or physical address)
- **6 bytes** long, typically written as **HEX**
- first three bytes identify a manufacturer (managed by IEEE)
- last three bytes determined by manufacturer (to produce unique addresses)
- Adapters normally only accept frames for their MAC and discard others.
- Adapters in promiscuous mode reads all frames
- The broadcast FF-FF-FF-FF-FF-FF is read by everyone



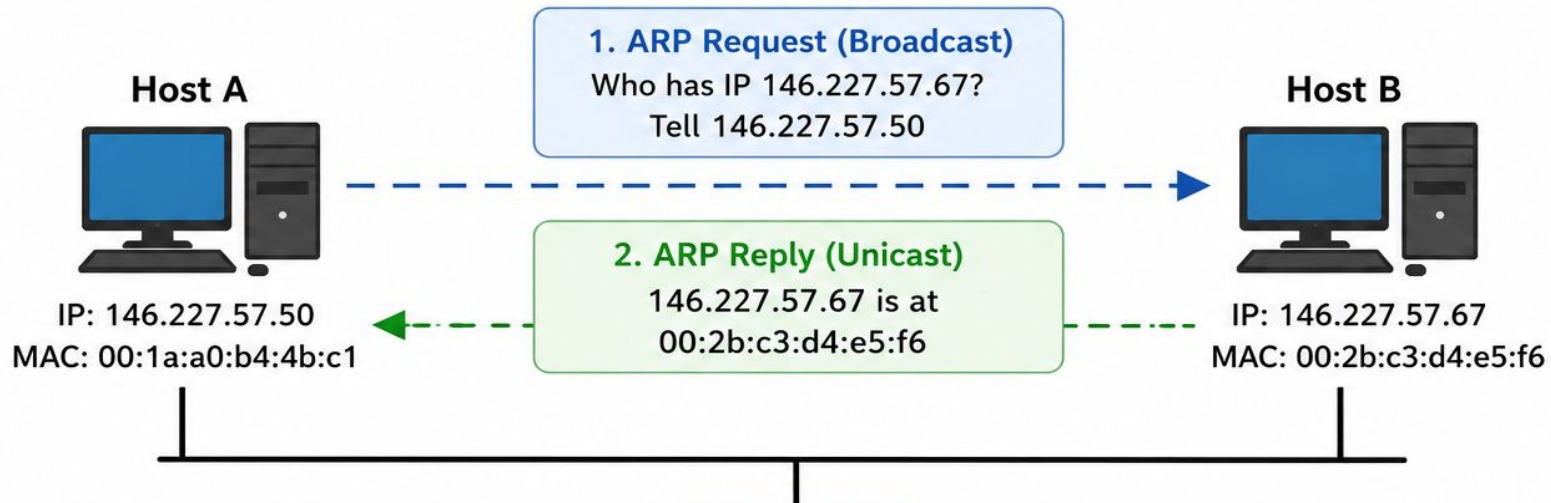


Address Resolution Protocol (ARP)

- ARP resolves an IP address to a MAC address
 - Link Layer: MAC eg 00:1a:a0:b4:4b:c1
 - Network Layer: IP eg 146.227.57.67
- ARP resolves an IP (e.g. 146.227.57.67) to a MAC address (e.g. 00:1a:a0:b4:4b:c1) in the same subnet

Address Resolution Protocol (ARP)

- ARP resolves an IP address to a MAC address
 - **Link Layer:** MAC eg 00:1a:a0:b4:4b:c1
 - **Network Layer:** IP eg 146.227.57.67
- ARP resolves an IP (e.g. 146.227.57.67) to a MAC address (e.g. 00:1a:a0:b4:4b:c1) in the same subnet



ARP Table

- Each host and router maintains an ARP table that contains mappings from IP to MAC
- Each entry has an expiry time

```
cfi@gh5: /proc/net
```

IP address	HW type	Flags	HW address	Mask	Device
10.13.51.77	0x1	0x2	48:ba:4e:f4:4a:fe	*	enol
10.13.51.254	0x1	0x2	cc:3e:5f:20:32:3b	*	enol
10.13.51.175	0x1	0x2	00:25:ab:89:c1:c6	*	enol

```
cfi@gh5: /proc/net$
```



ARP continued

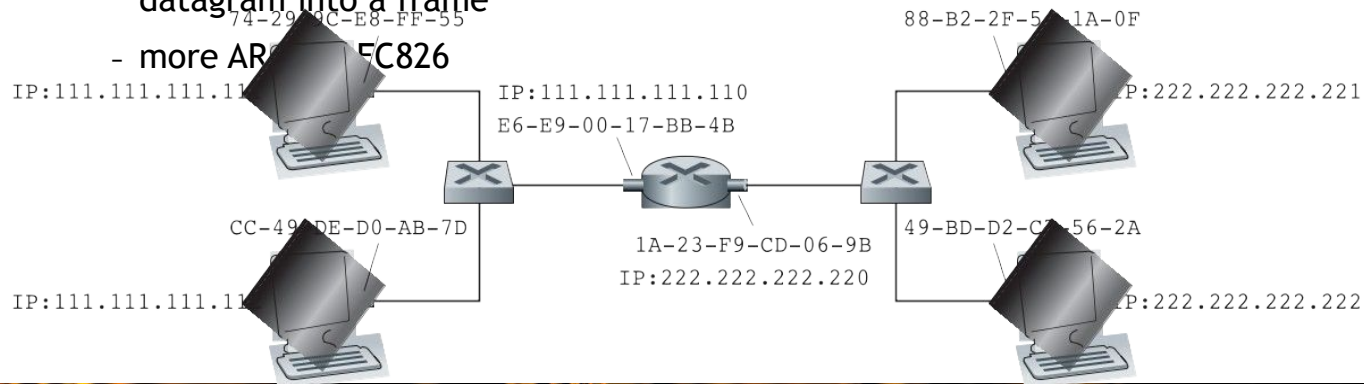
A host sending a datagram to another host/router in the subnet needs to resolve the IP to a MAC address

- If the ARP table contains the MAC address entry it can assemble the frame and transmit
- If the ARP table does not contain the address ARP protocol is used to discover it:
 1. Sender constructs a special **ARP packet**
 - contains header fields that include the sending and receiving IP addresses and the MAC addresses
 - both query and response packets have the same format.
 2. Sender encapsulates the ARP query packet in a frame and transmits to **FF-FF-FF-FF-FF-FF**
arp 42 who has 192.168.1.1? tell 192.168.1.33
 3. Frame is received by all NICs on the subnet and passed to their ARP module
 4. Each NIC checks if the requested IP address is used by that host
 - If yes, then the NIC replies with the ARP response in a standard frame back to the query IP/MAC
 - If no the ARP query is discarded (or in some cases the IP/MAC is cached in the ARP table)
- Now the sender can update the ARP table and send the datagram



ARP to a remote subnet

- What happens if the datagram needs to be sent off the current subnet?
- Router has an IP/MAC for each of its interfaces
 - (subnets 111.111.111/24 and 222.222.222/24)
- A sender in subnet 1 would use ARP to discover the router
111.111.111.110 with MAC E6-E9-00-17-BB-4B
 - router determines the outgoing interface using a forwarding table
- The outgoing interface (222.222.222.220) passes then the datagram to subnet 2
 - using ARP to discover the destination's MAC before putting the datagram into a frame



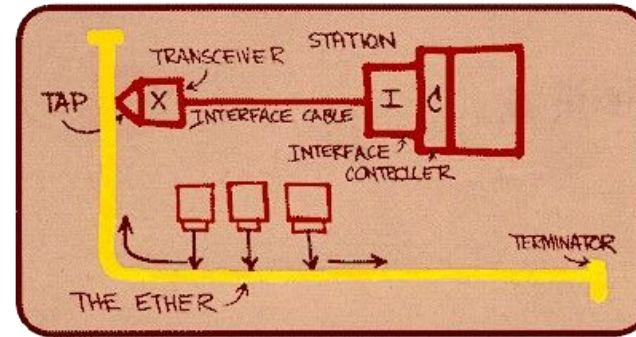
Lecture Content

- Principles of Link Layer services
- Link Layer protocols
- Link Layer Addressing
- **Ethernet Protocol**
- Switches and Hubs



Ethernet

- Dominant wired LAN technology:
 - Cheap @ < £10 for basic NIC
 - kept up with speed race: 10 Mbps - 10 Gbps
 - 400 Gbps soon
 - Terabit Ethernet
 - >1000Gbps

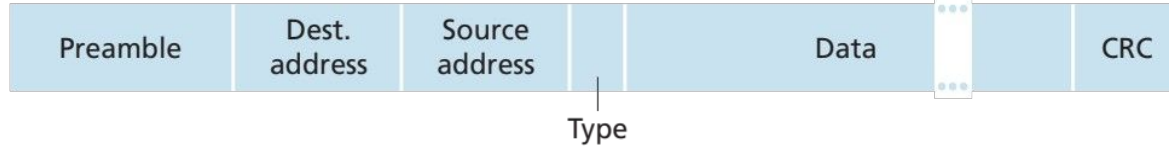


*Metcalfe's Ethernet
sketch*



Ethernet Frames

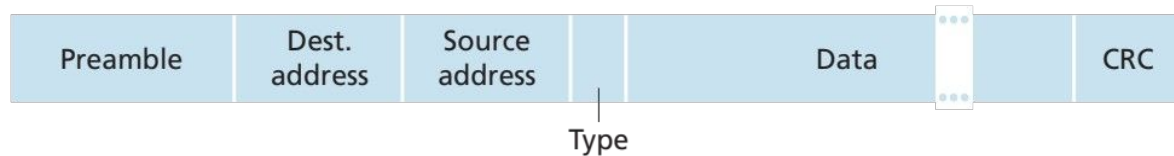
- **Preamble (8 bytes):**
 - AA AA AA AA AA AA AB (used as a wakeup call and for clock synchronisation)
 - The last two bits (...11) are a marker
- **Destination Address (6 bytes)**
 - The MAC address of the destination adaptor
- **Source Address (6 bytes)**
 - The MAC address of the transmitting adaptor





Ethernet Frames

- **Type** (2 bytes)
 - value ≤ 1500
 - . size of payload in octets
 - Value ≥ 1536
 - . EtherType - defines which protocol is encapsulated in the payload of the frame
- **Data** (46 - 1,500 bytes)
 - the IP datagram
 - . If an IP datagram is larger than 1,500 bytes, it needs to be “fragmented” (network layer)
 - . If it is less than 46 bytes it needs to be “stuffed”.
- **CRC** (4 bytes)
 - Detect errors in the frame





Ethernet Services

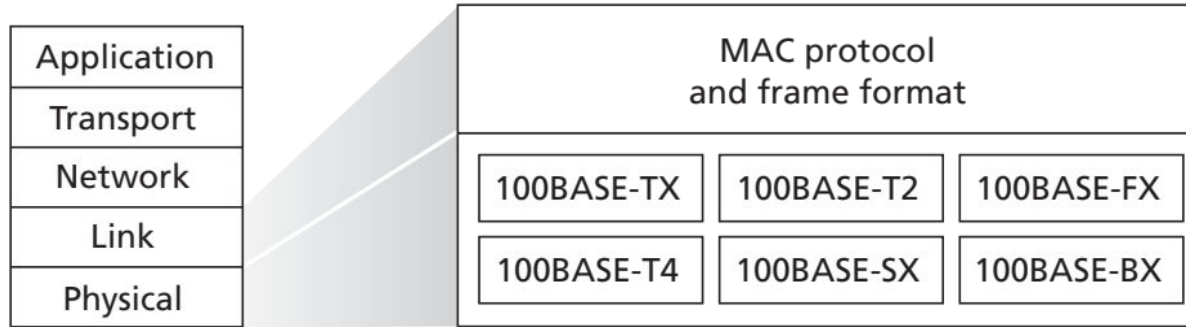
- Ethernet provides a **connectionless** service to the network layer
- The ethernet service is “**unreliable**”
 - if a frame fails the CRC check it is simply discarded
 - no acknowledgement is sent

Ethernet Services

- Stream of datagrams (network layer) can have gaps
- These gaps are handled at the transport layer (TCP) or not (UDP)
- Ethernet's MAC protocol
 - unslotted CSMA/CS with binary backoff



Ethernet Standards: 802.3

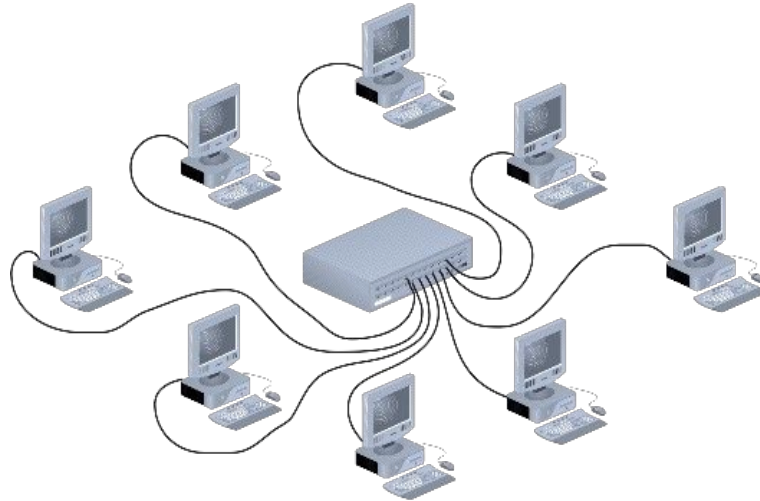


- Ethernet is more than Link Layer
 - also specifies the physical layer
- Common MAC protocol and frame format
- Different speeds and physical layer media
- For Gigabit Ethernet with modern full-duplex switches the MAC protocol is becoming obsolete



Ethernet: physical topology

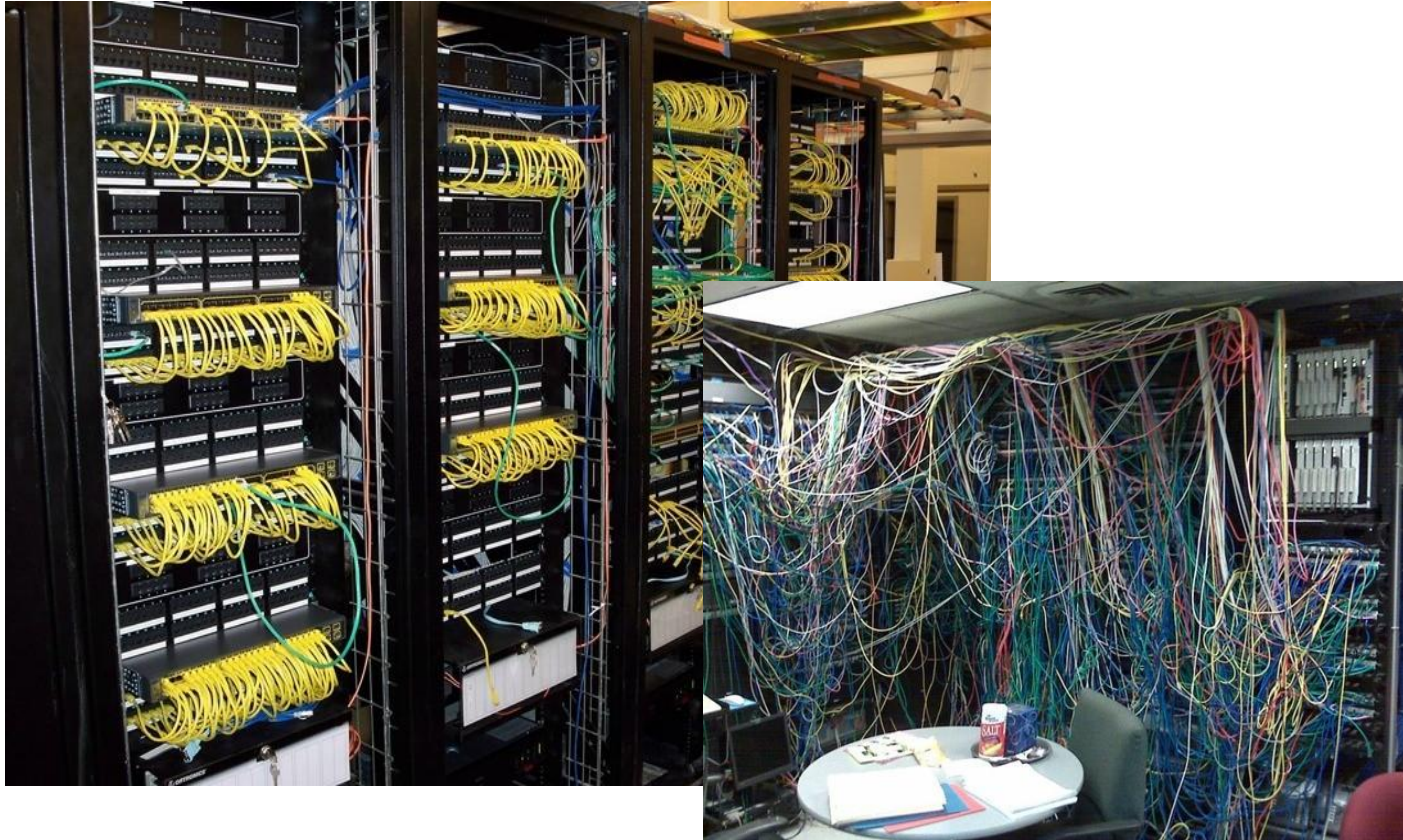
- Star
 - prevails today
 - active switch in centre
 - each “spoke” runs a (separate) Ethernet protocol
 - nodes do not collide with each other)



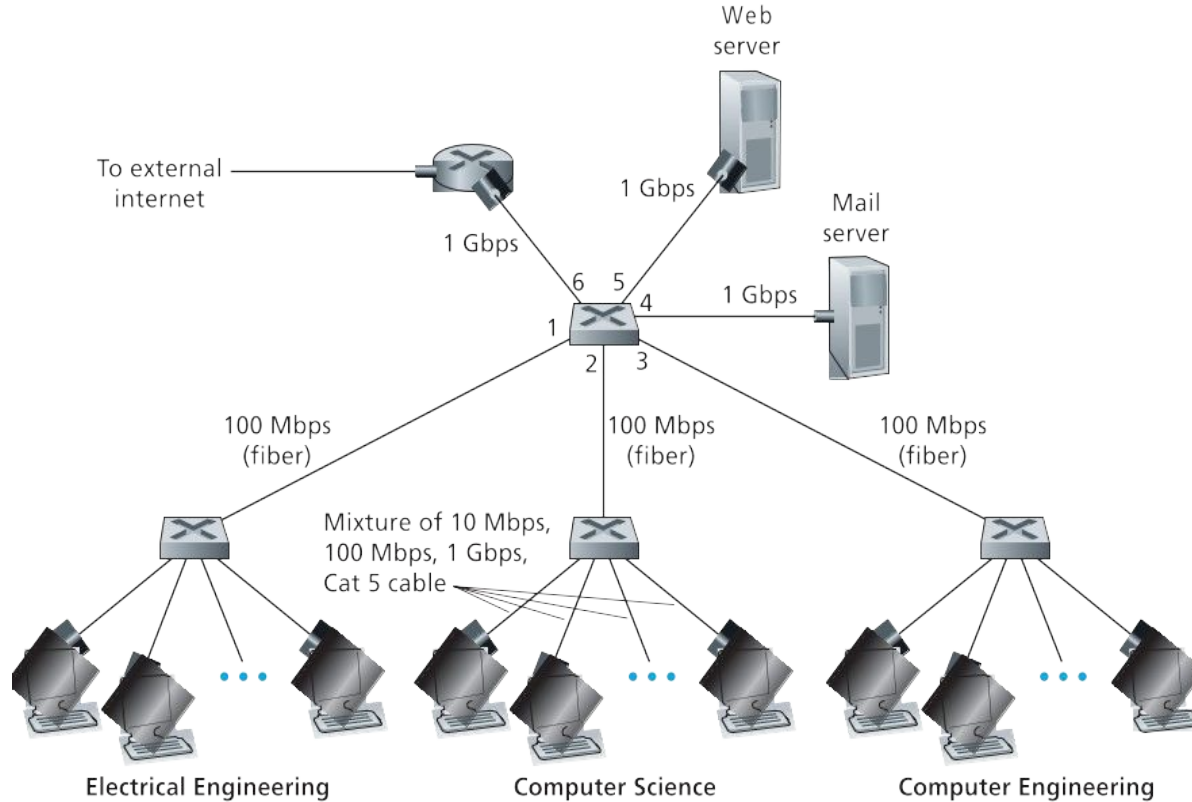
Lecture Content

- Principles of Link Layer services
- Link Layer protocols
- Link Layer Addressing
- Ethernet Protocol
- **Switches and Hubs**

Switch (cab) room



Link Layer Switches





Link Layer Switches

- Connect networks
- Operate on Link Layer addresses
 - NOT IP addresses
- Remember:
 - Each NIC has a link-layer address
- Switches are transparent
 - they do not have a link layer address



Switches: Filtering and Forwarding

- Filtering determines whether a frame should be dropped
- Forwarding determines the interface to which a frame should be directed and moves the frame to that interface



Switches: Filtering and Forwarding

- This is how the switch decides:
 - If there is no table entry it forwards the frame to all other interfaces
 - If there is an entry and the entry is for the same interface from which the frame originates
• the frame is filtered
 - If there is an entry and the entry is for another interface than the one the frame originates, then the frame is forwarded to that interface

Address	Interface	Time
62-FE-F7-11-89-A3	1	9:32
7C-BA-B2-B4-91-10	3	9:36
....		



Switches: Self-learning

- The switch learns about connected interfaces
 - Initially the switch table is empty
 - For each incoming frame the MAC address from the frame's source address field, the interface on which it arrives and the current time are stored
 - an entry is removed from the switch table after some period of time (Aging Timer)
- No need for manual configuration -> switches are plug and play devices

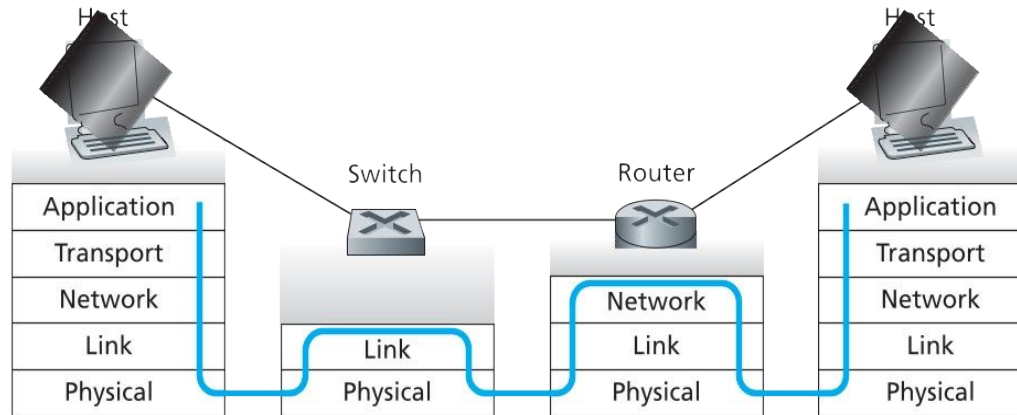


Link Layer Switches vs Hubs

- A switch learns which MAC is connected to which socket
 - and forwards the frame
- A hub broadcasts a frame to all other ports

Switches vs Routers

- both are store-and-forward:
 - routers: network-layer devices (examine network-layer headers)
 - switches: link-layer devices (examine link-layer headers)
- both have forwarding tables:
 - routers: compute tables using routing algorithms, IP addresses
 - switches: learn forwarding table using flooding, learning, MAC addresses



Lecture Content

- Principles of Link Layer services
- Link Layer protocols
- Link Layer Addressing
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- Switches and Hubs



http://global.dlink.com.sg/site_img/Products/DES-3200-Series/DES-3200-main.jpg



Thank You!